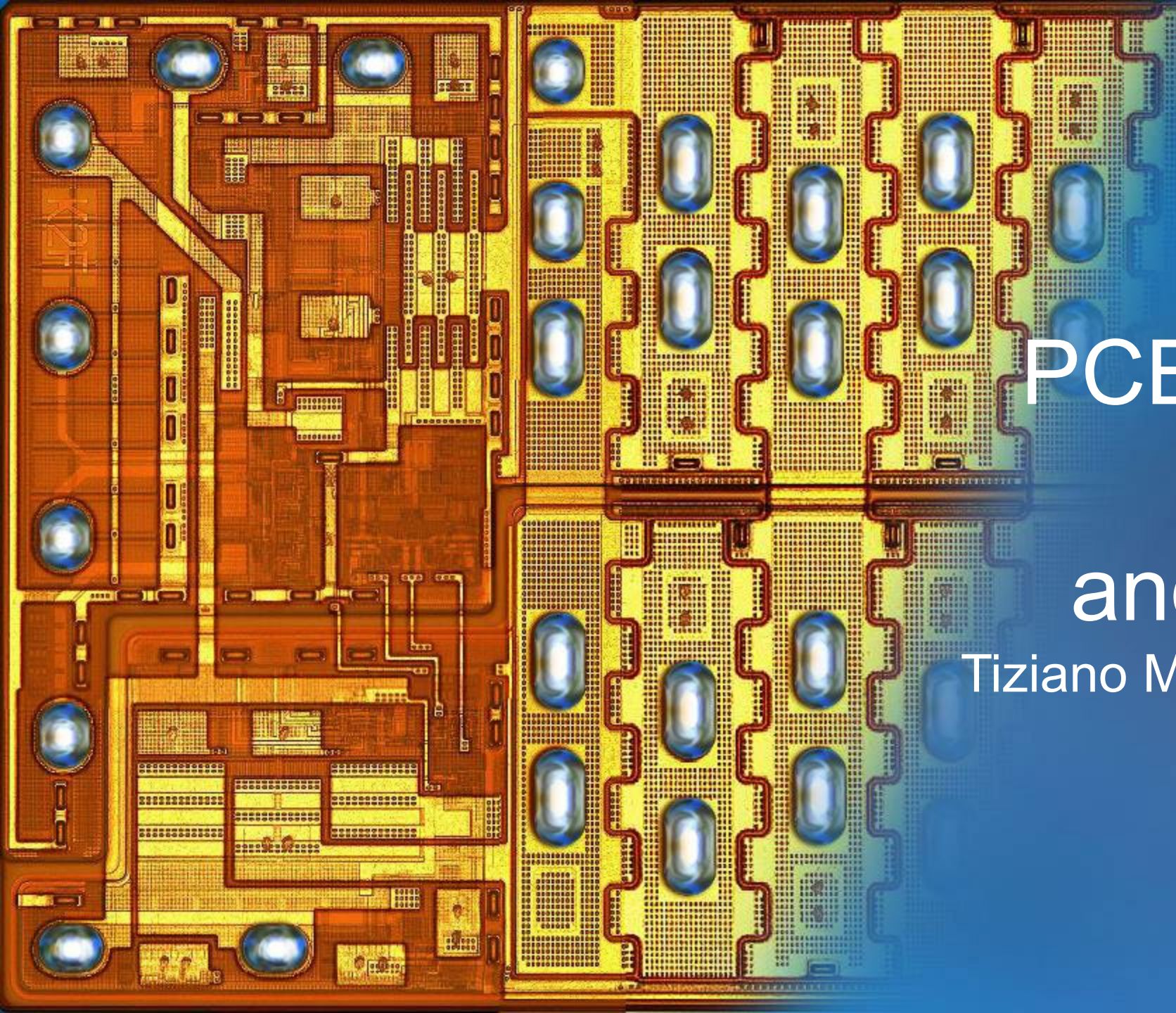


PCB Layout Rules for Converters and Motor Drives

Tiziano Morganti – April 29th, 2025

EFFICIENT POWER CONVERSION

Overview

- Parasitic Inductances and their Impact
- Layout Design Comparisons
- Designing a Low Inductance Layout
- Gate Circuit Layout
- Common Mistakes Example and Fixes
- Parallel Design Example
- PCB Properties and Requirements for EPC FETs
- Motor Drive vs DC/DCs

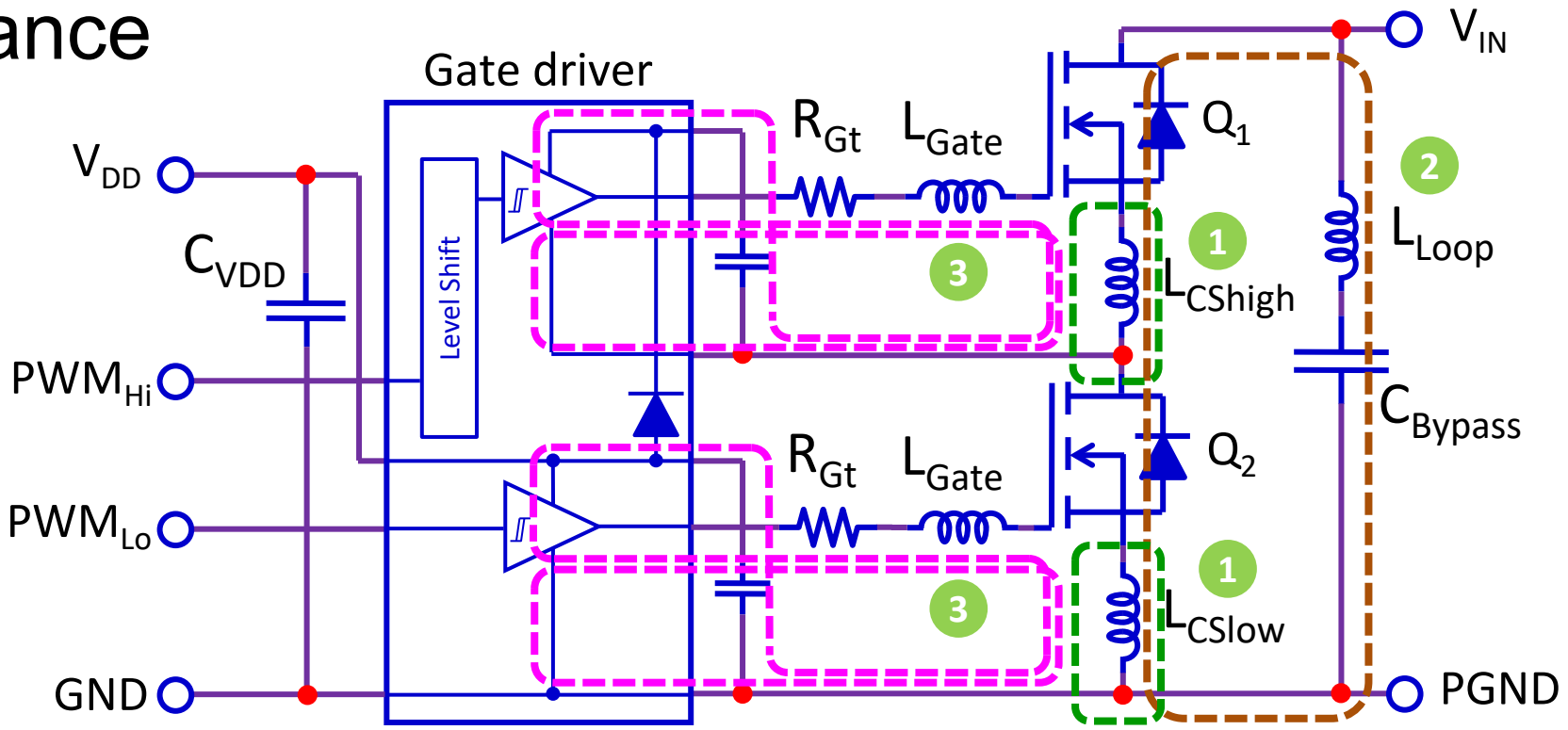
Parasitic inductances

1. Common source inductance
2. Power loop inductance
3. Gate loop inductance
   - Turn-on
   - Turn-off

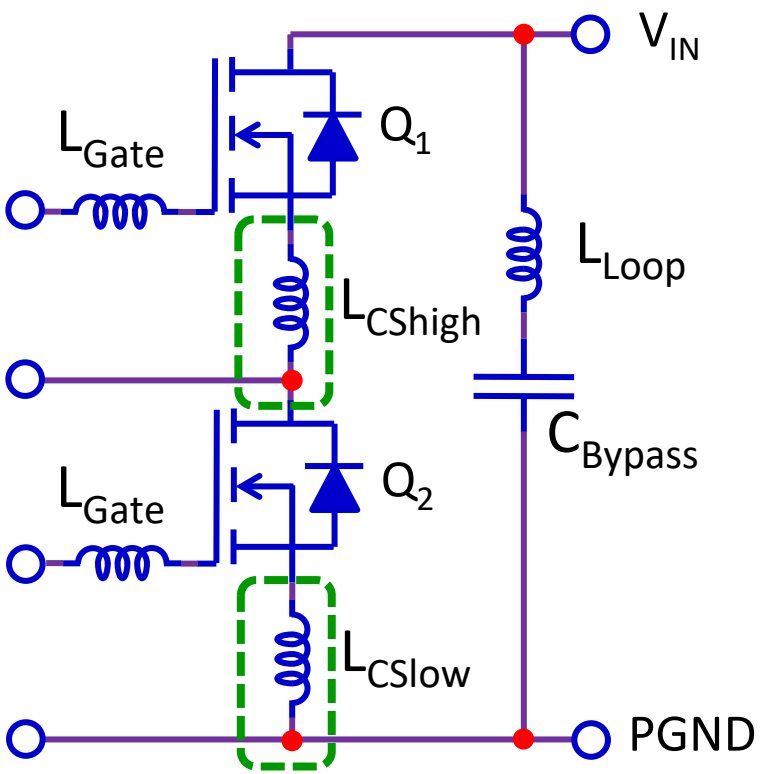
[WP010: Optimizing PCB Layout](#)

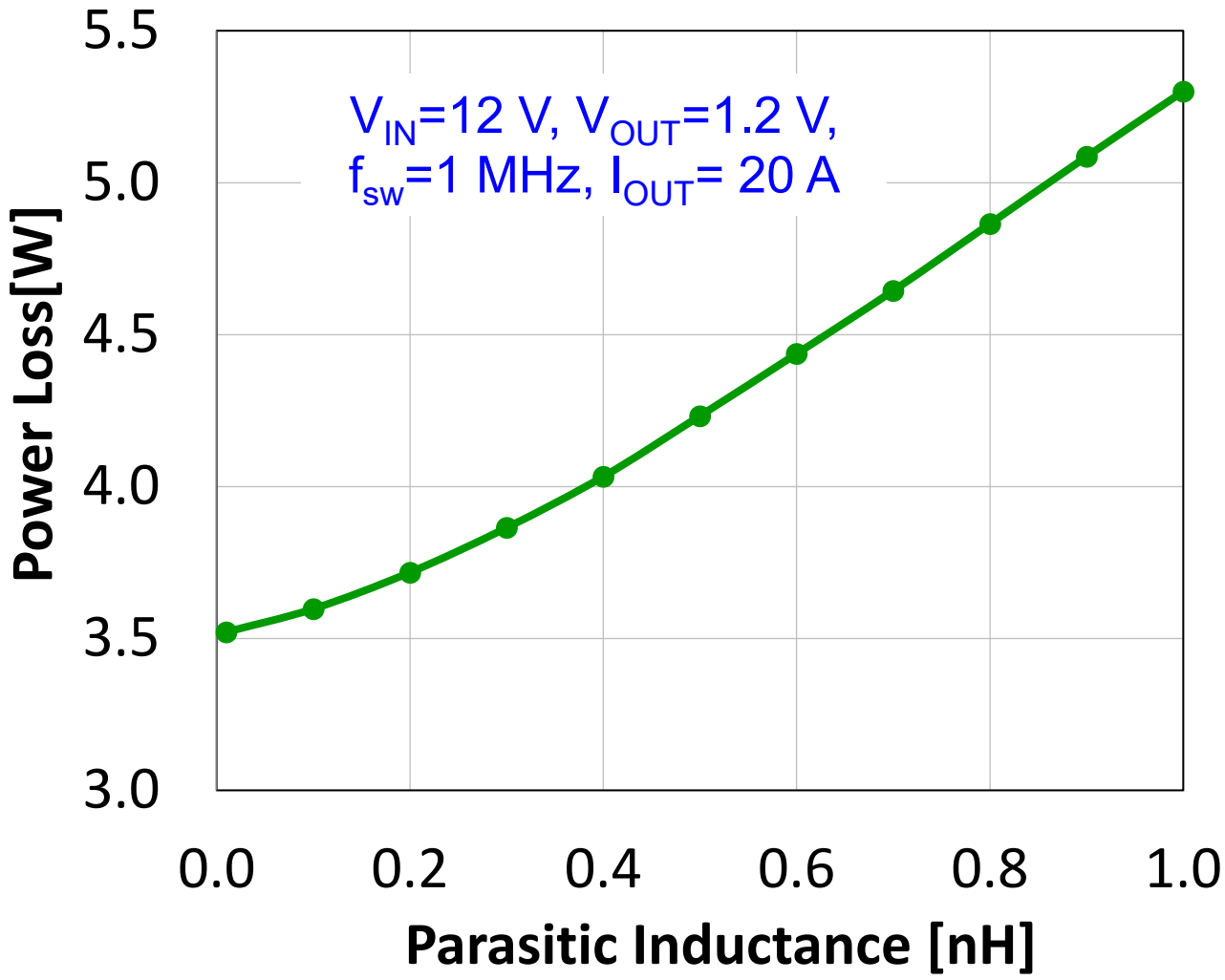
[How to GaN 05](#)

[How to GaN 05a](#)

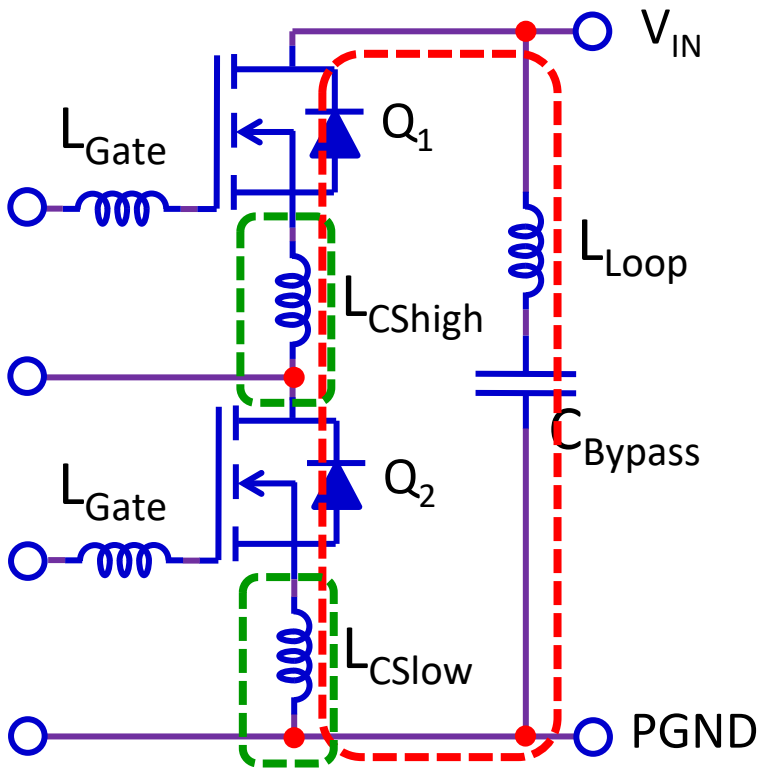


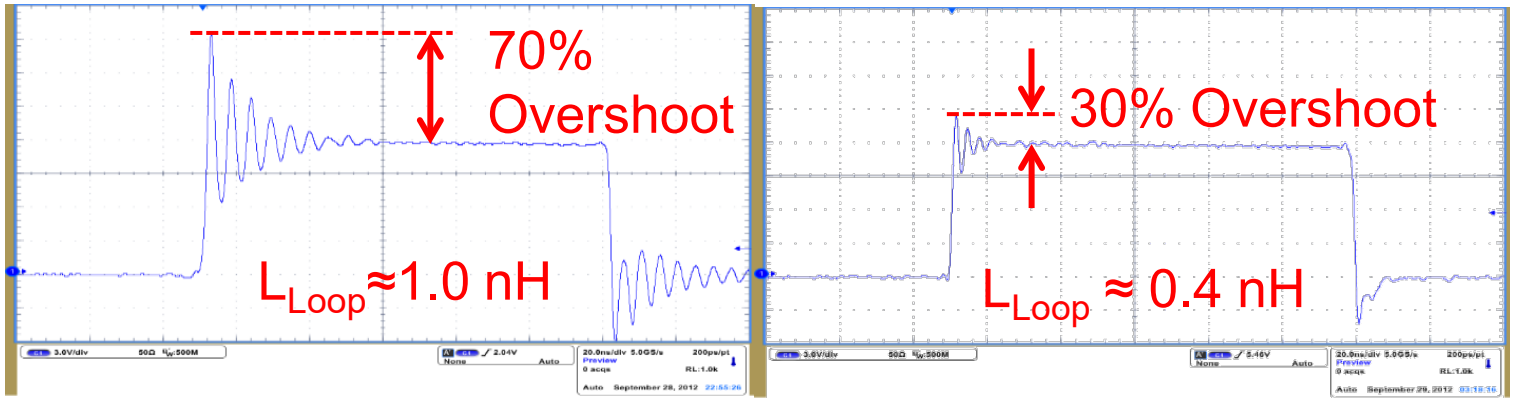
Impact of Common Source Inductance

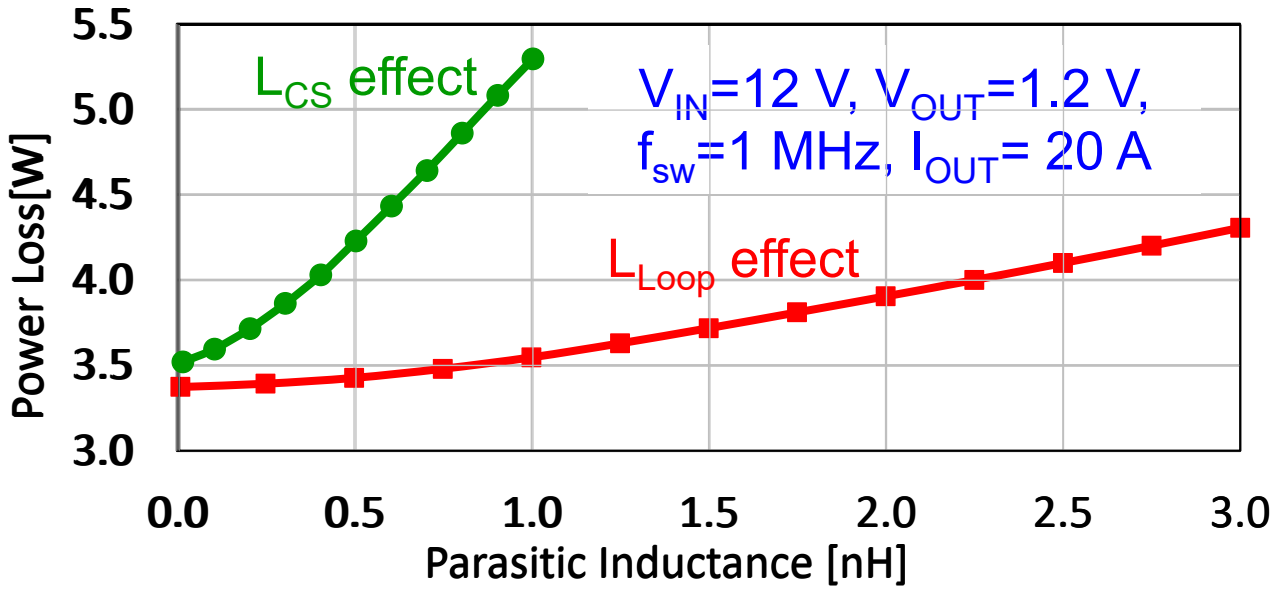




Impact of Power Loop Inductance

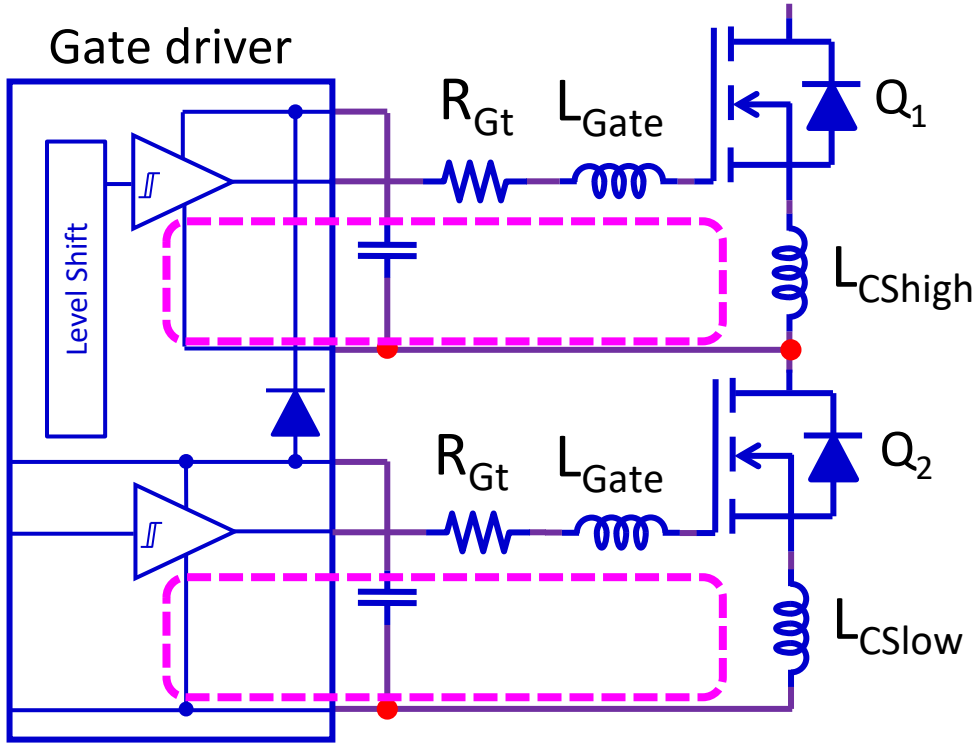




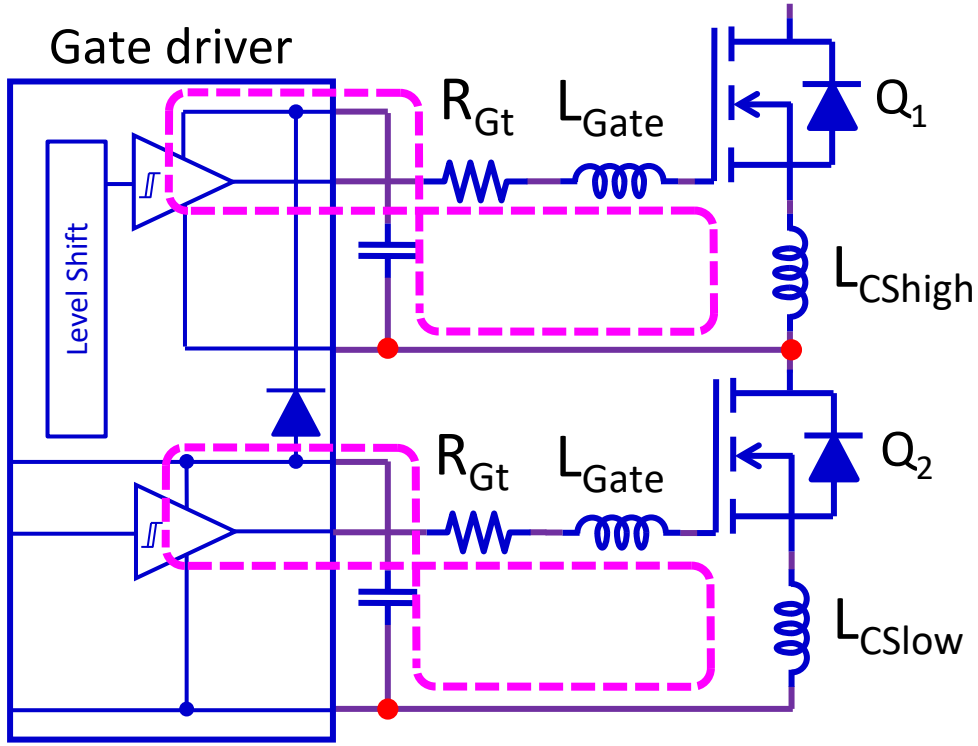


Impact of Gate Loop Inductance

- Two loops to consider: Turn-on & Turn-off
- L_{Gate} requires R_{Gt} to damp ringing overshoot
  - slows transition

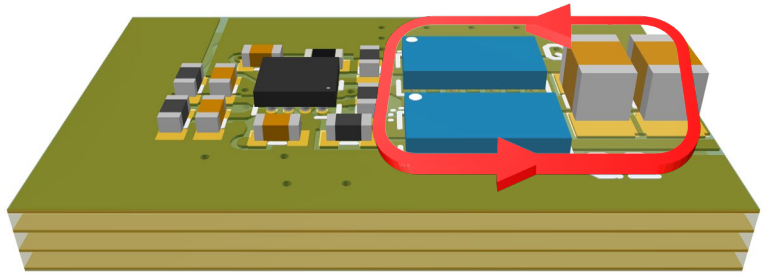


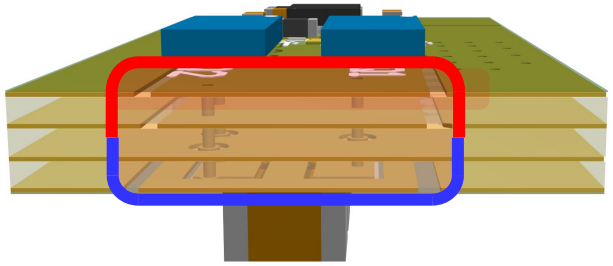
Turn Off

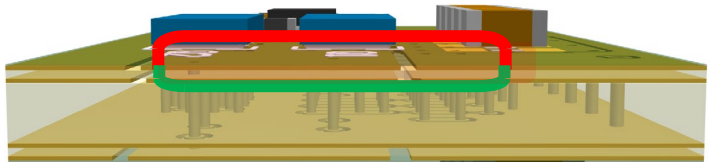


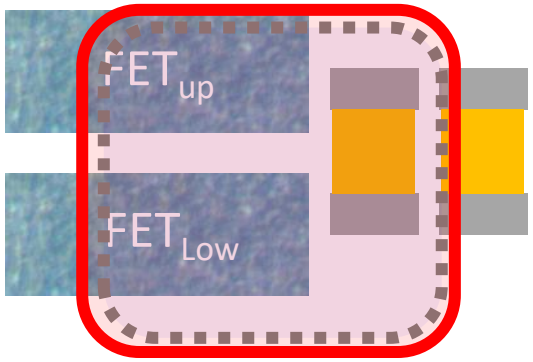
Turn On

Layout Comparisons



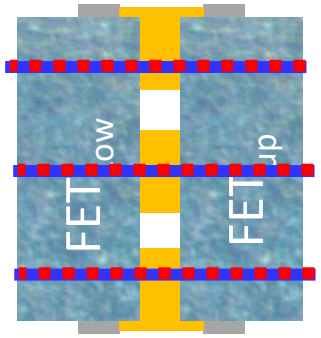






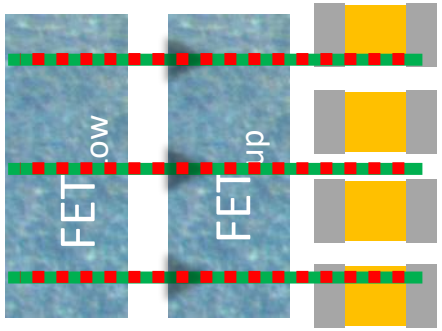
Top Layer

Mirror (Inner Layer 1)



Top Layer

Bottom Layer



Top Layer

Inner Layer 1

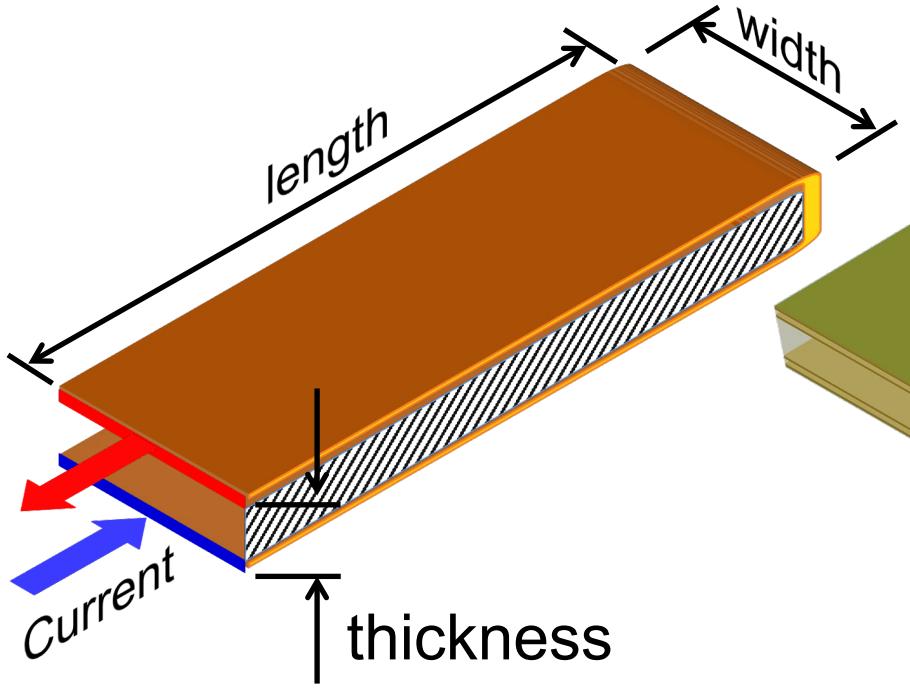
Lateral

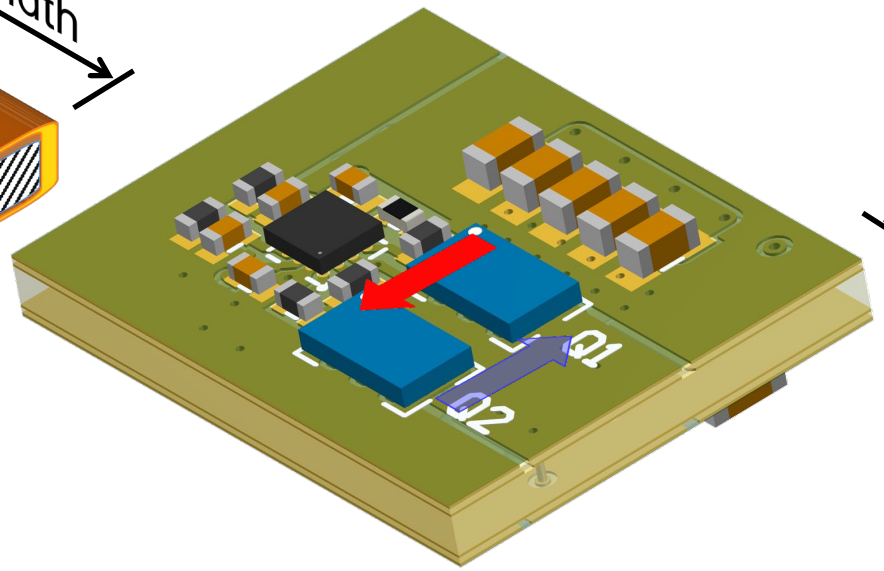
External Vertical

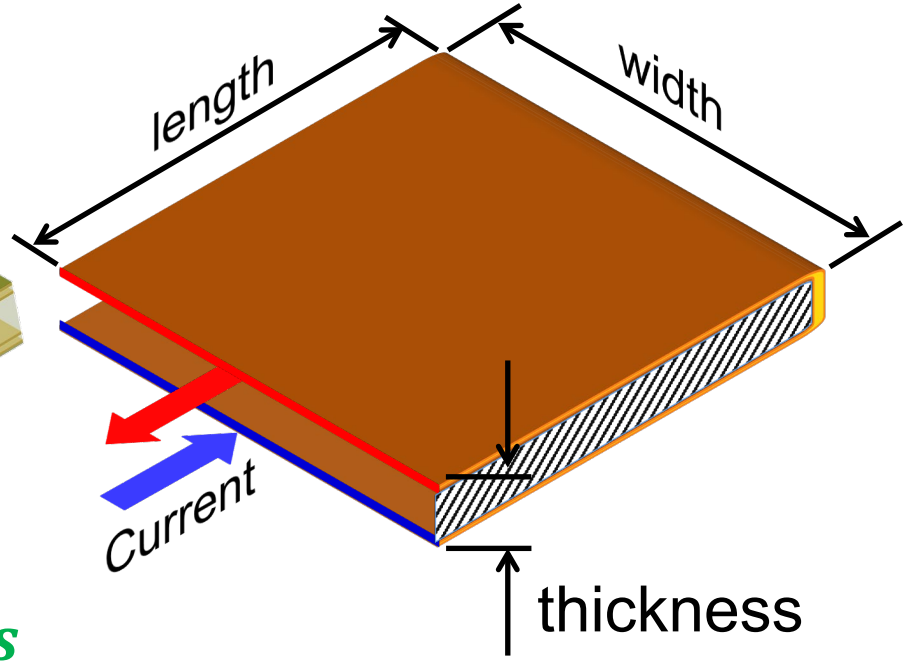
Internal Vertical

How Inductance Manifests in a Layout

Which layout has higher inductance?







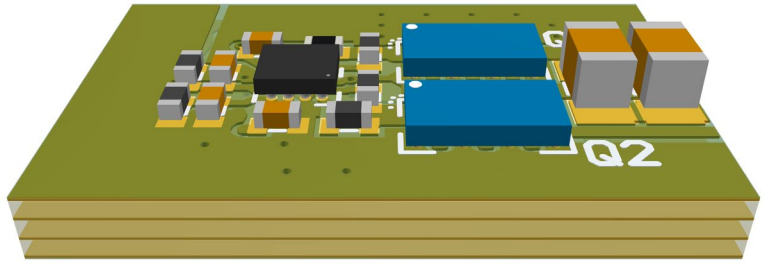
$$L = \frac{\mu_0 \cdot length \cdot thickness}{width}$$

Long, Narrow

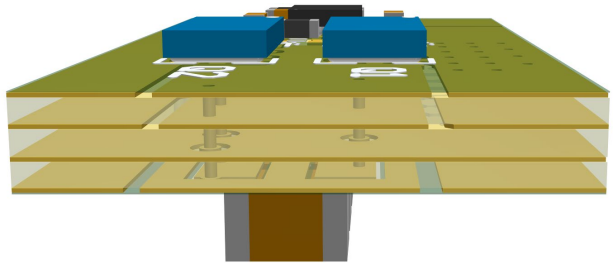
Short, Wide

$$L_{LN} > L_{SW}$$

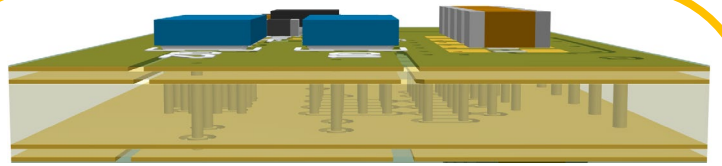
Layout Inductance Comparison



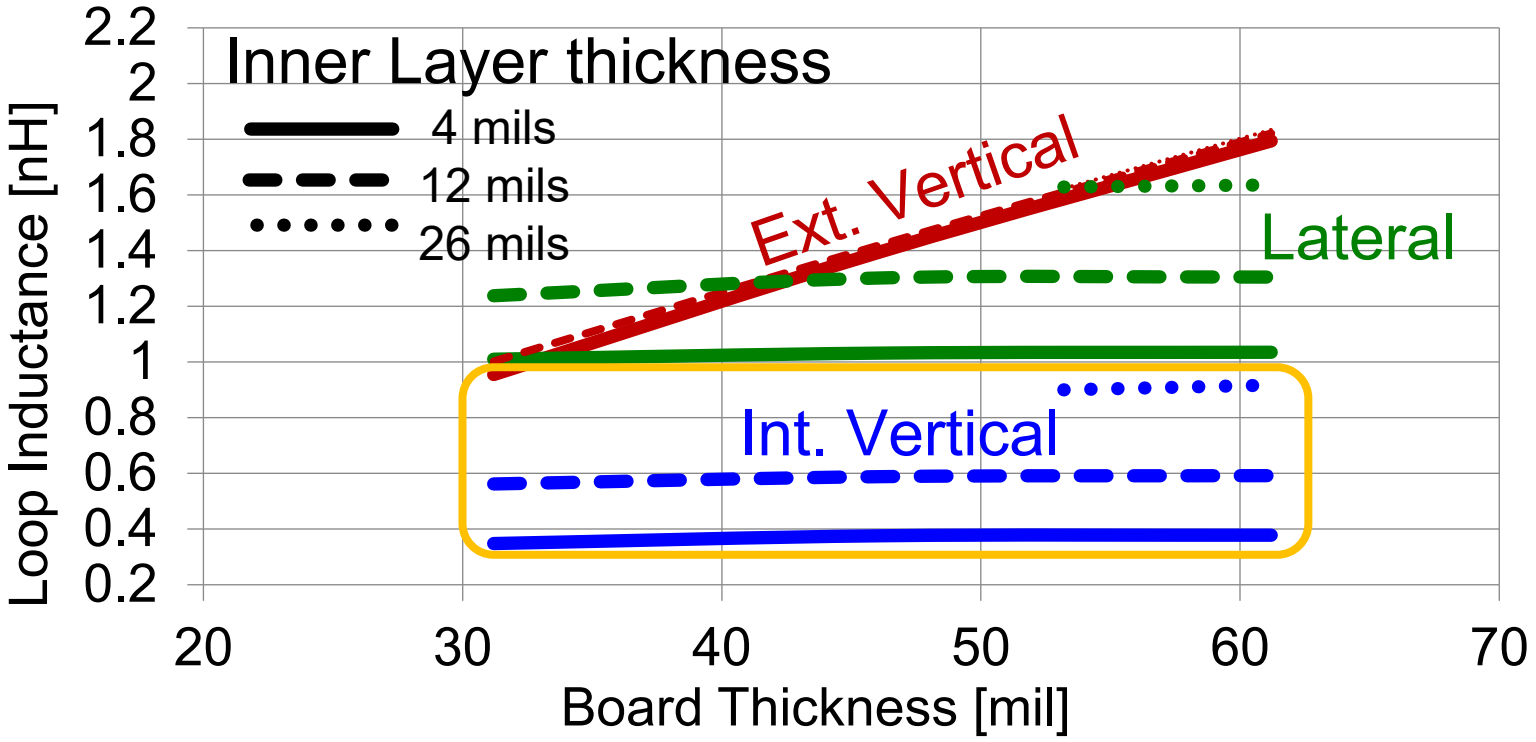
Lateral

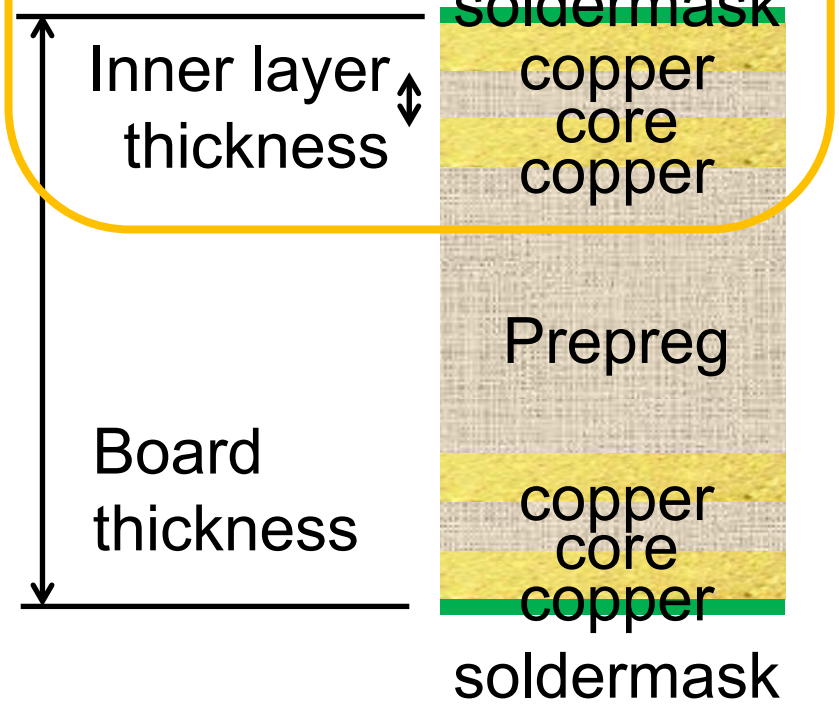


Ext. Vertical



Int. Vertical
Best Choice

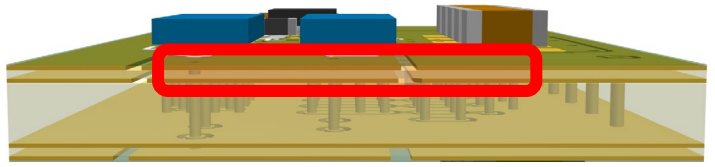


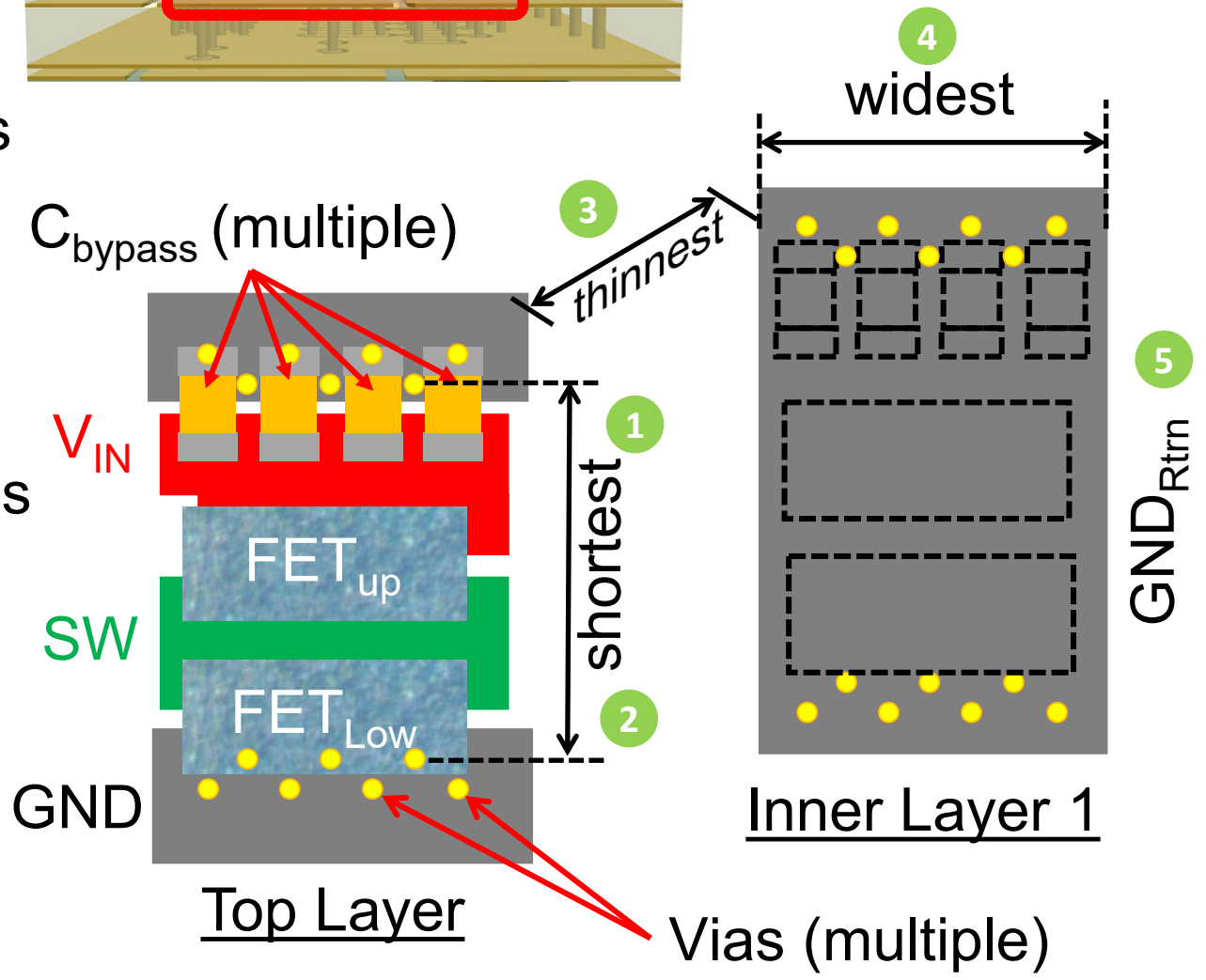


Designing a Low Inductance Layout

Internal vertical (optimal) power loop:

1. Vertically place components as close as possible
2. Place (many) vias as close to the innermost electrical connection
3. Thinnest permissible substrate thickness between outer and first inner layer
4. Spread out via connections at innermost connect
5. GND return does **not** need to carry the full current

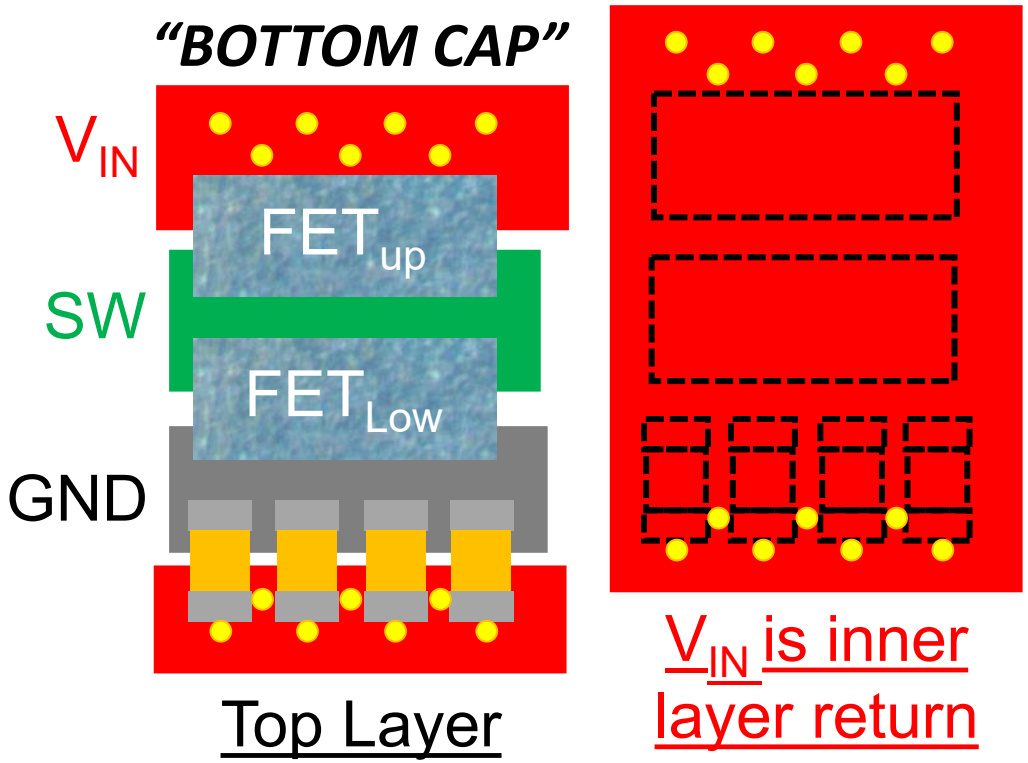




Alternative Layout Configurations

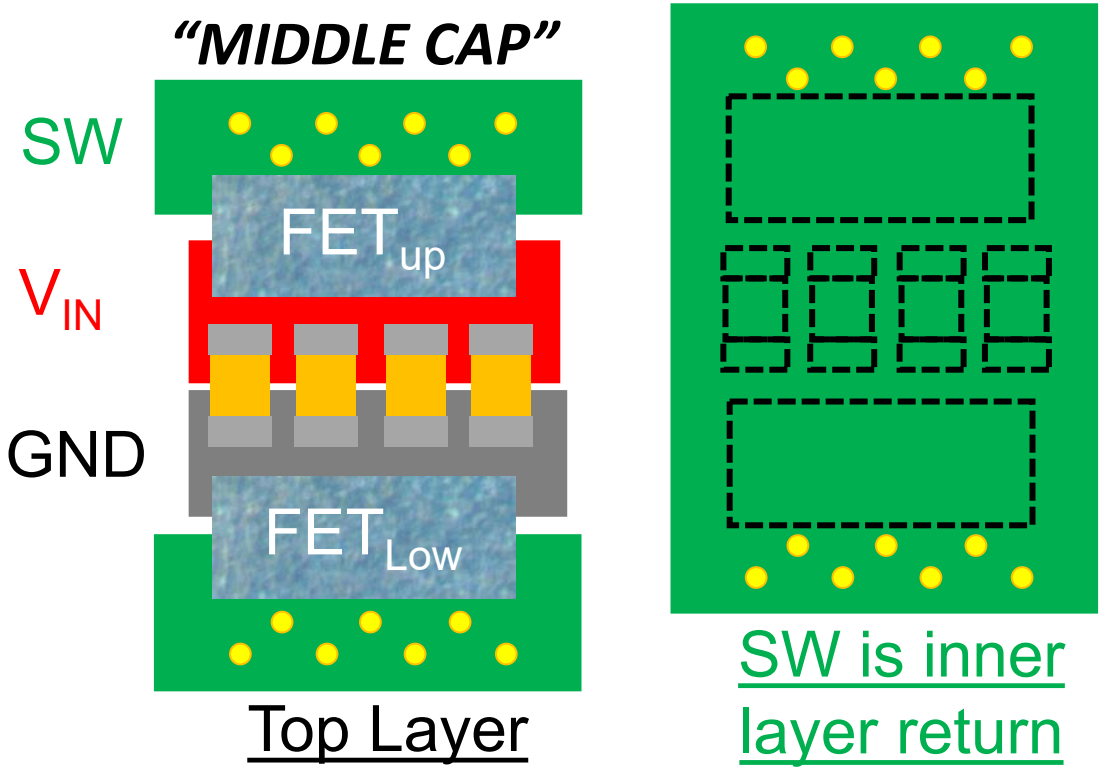
Bus capacitors beside lower FET

- Best cooling for upper FET



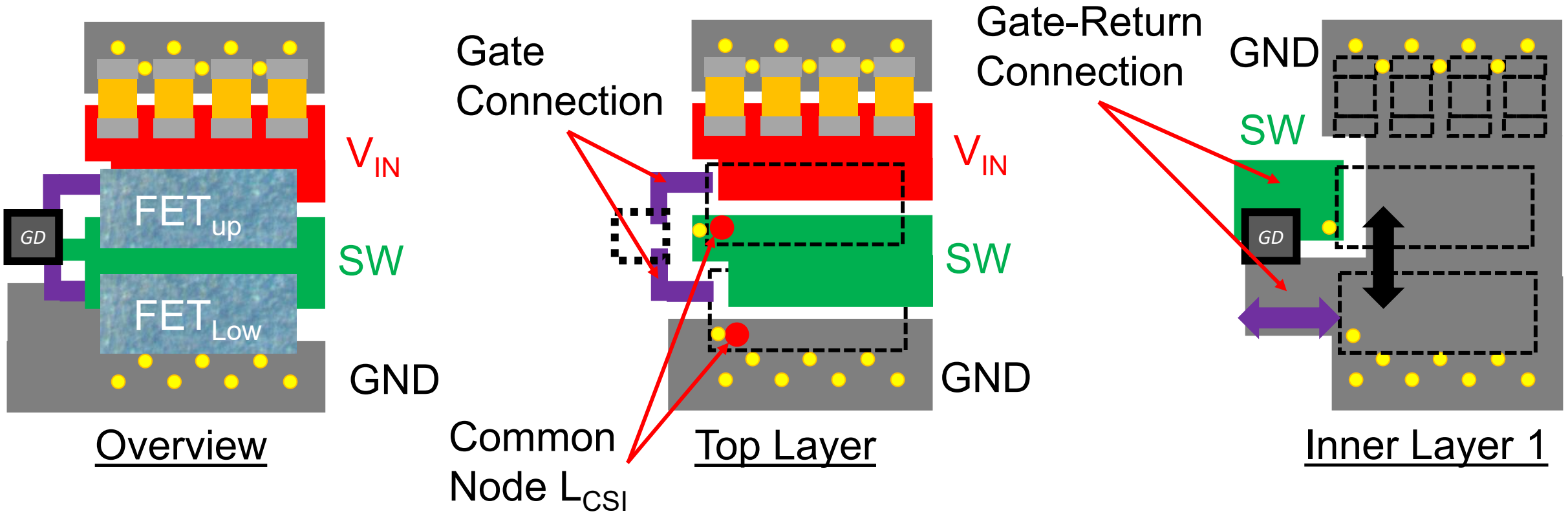
Bus capacitors between the FETs

- Spread the heat apart
- Buried switch node



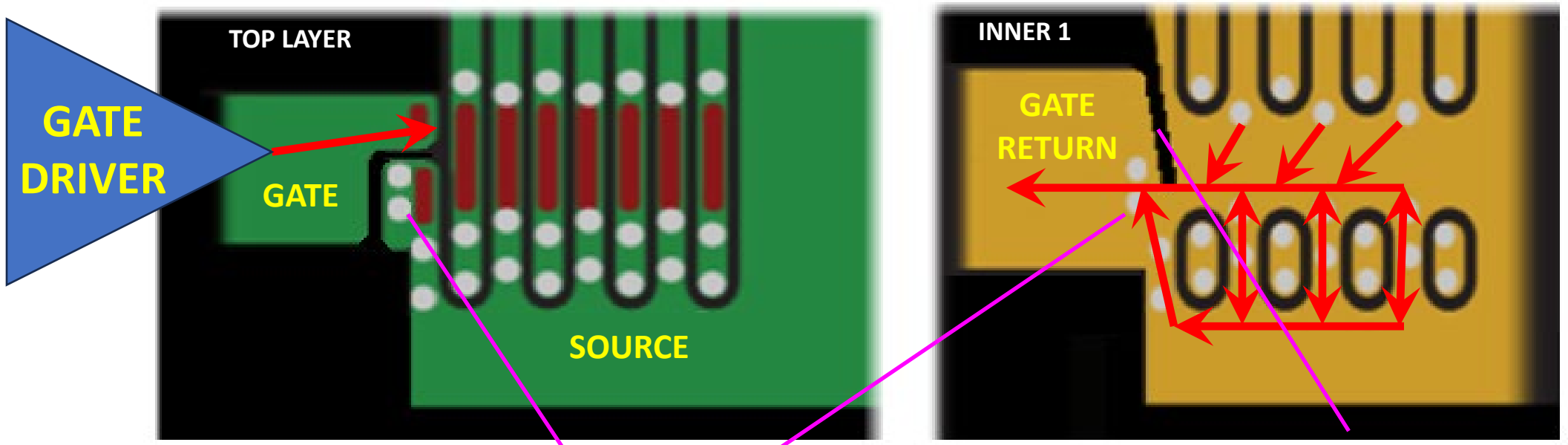
What About the Gate Connection

- Gate drive track with return polygon (its source!) on next layer
- Orthogonal gate to power connection reduces coupling between them



To Kelvin or not to Kelvin?

- Connect **ALL** the source pads together
- place one or few vias near to the source pad closest to the gate pad
- those vias will be the only connection between power loop and gate return polygon: this is the Kelvin connection

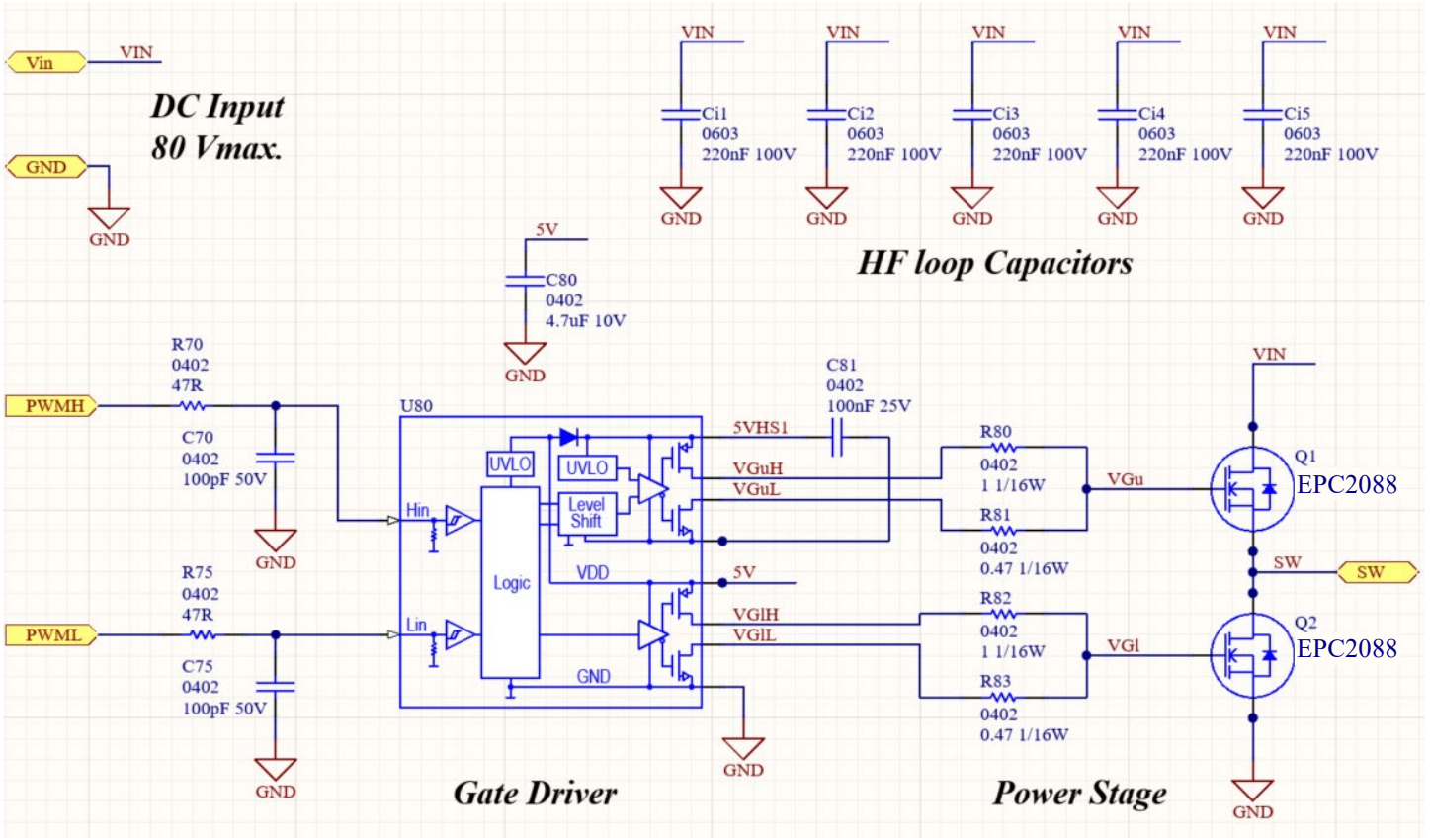


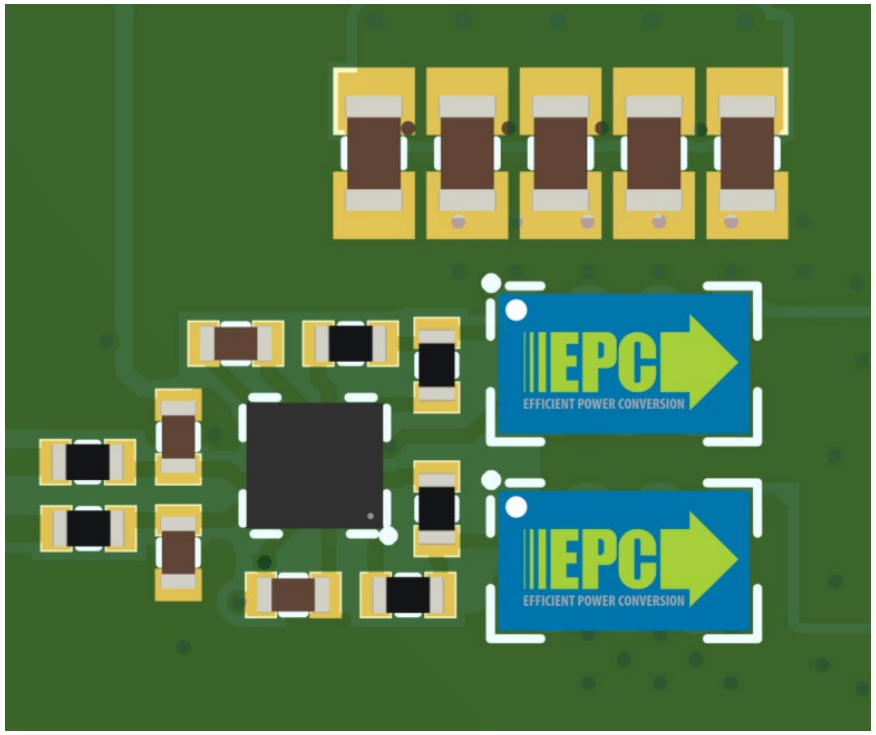
Final tips

- Prioritize the switch gate layout (HS in buck, LS in boost)
- Use smaller capacitors (10 to 100 nF, 0603) to close the power loop:
  - they deal with switching transient only
- Then, larger capacitor placement is less critical:
  - they deal with ripple currents at F_{SW}
- Avoid unnecessary stuff in the power loop / gate drive
  - Prioritize components dealing with switching edges
- Take care of the component/footprint library
  - EPC offers Altium and Kicad libraries, and footprint drawings in datasheet
  - Silkscreen matters: avoid surrounding components

Case study

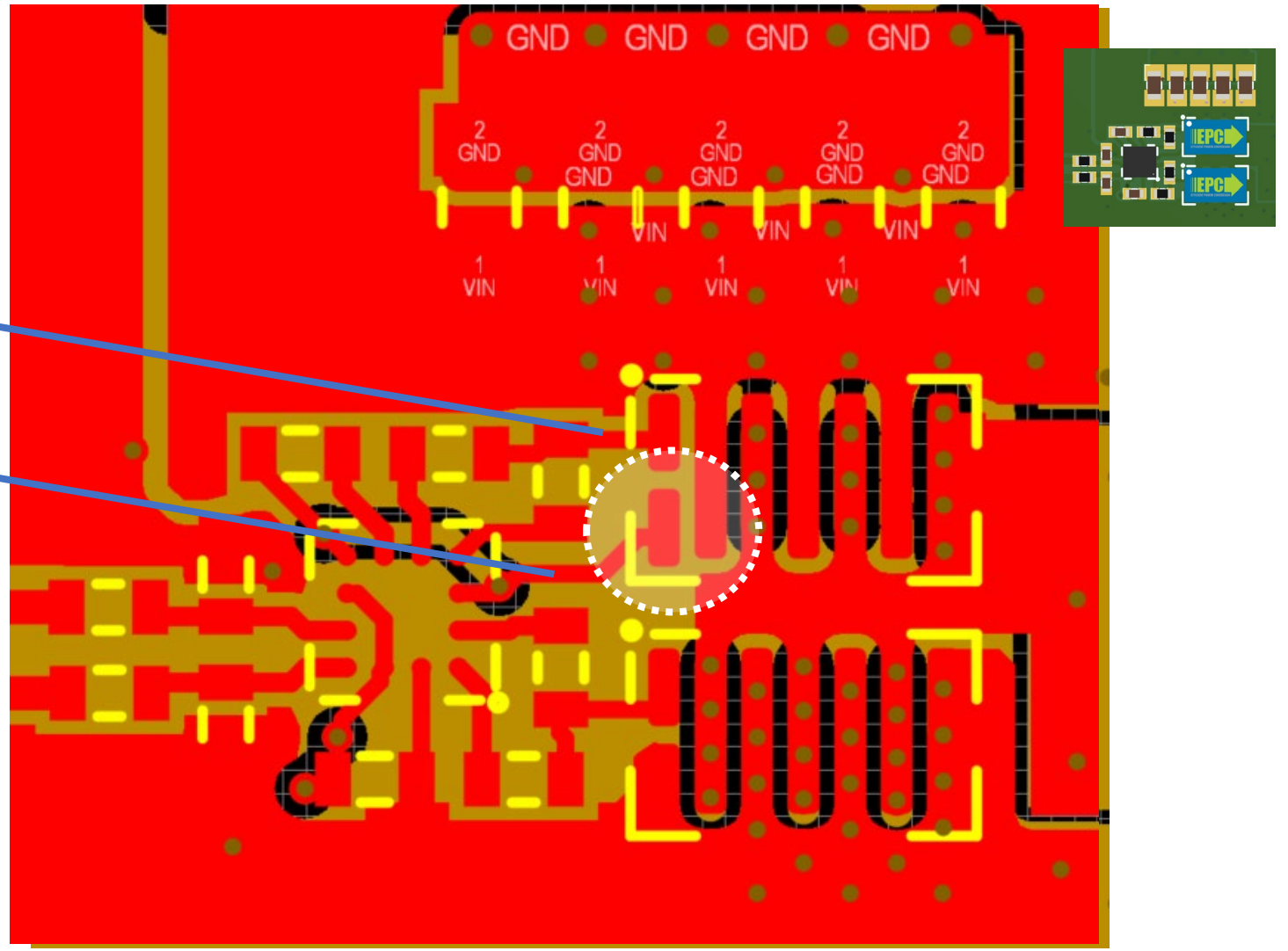
EPC2088 half bridge with LMG1205/MP8699B/LT8418





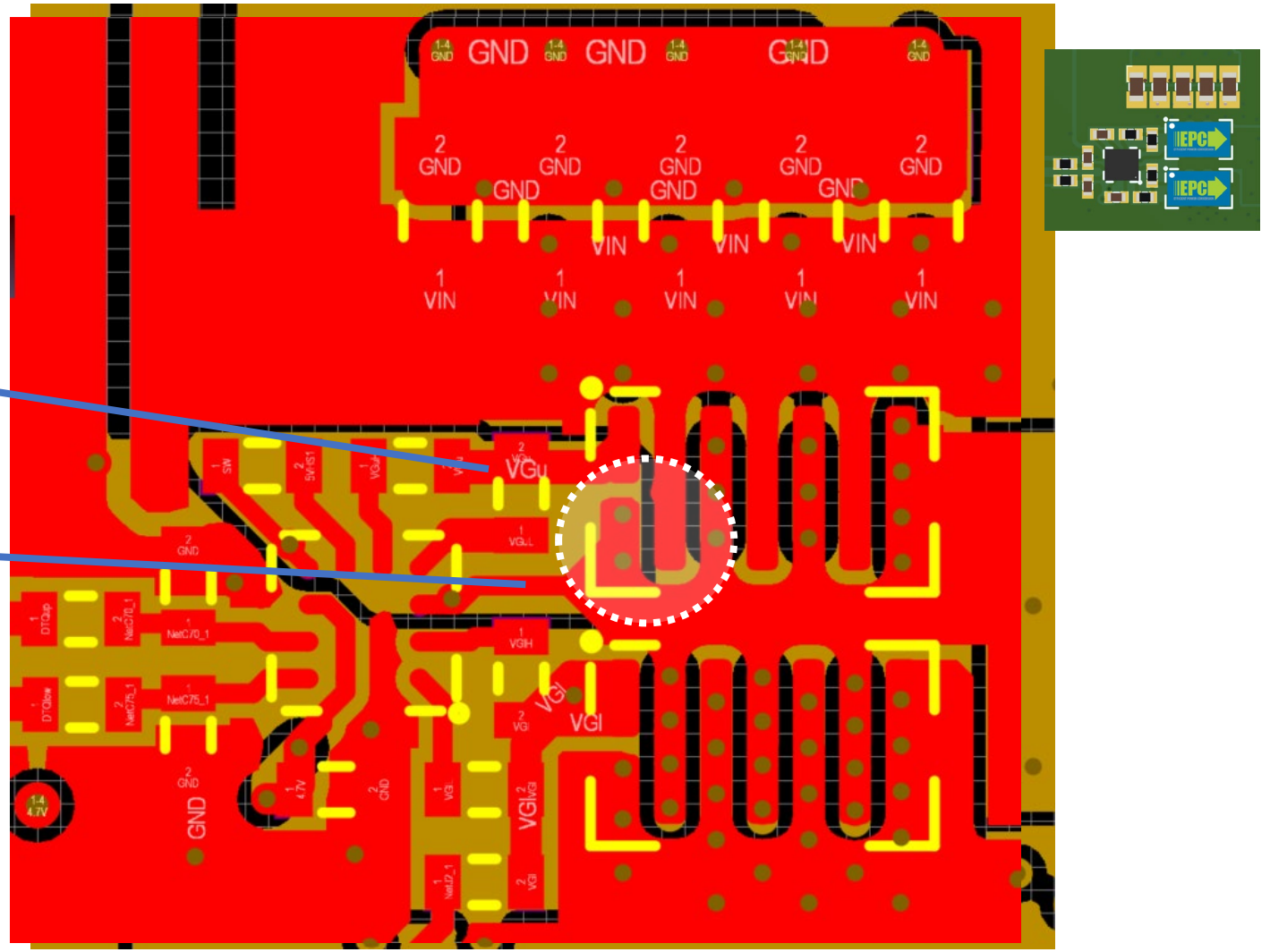
Common Mistakes in Gate Layout

- Pad 2 used only as gate return Kelvin connection
- Thin tracks for gate drive and return
- Gate drive return in top layer only
- HS floating gate circuitry over GND polygon



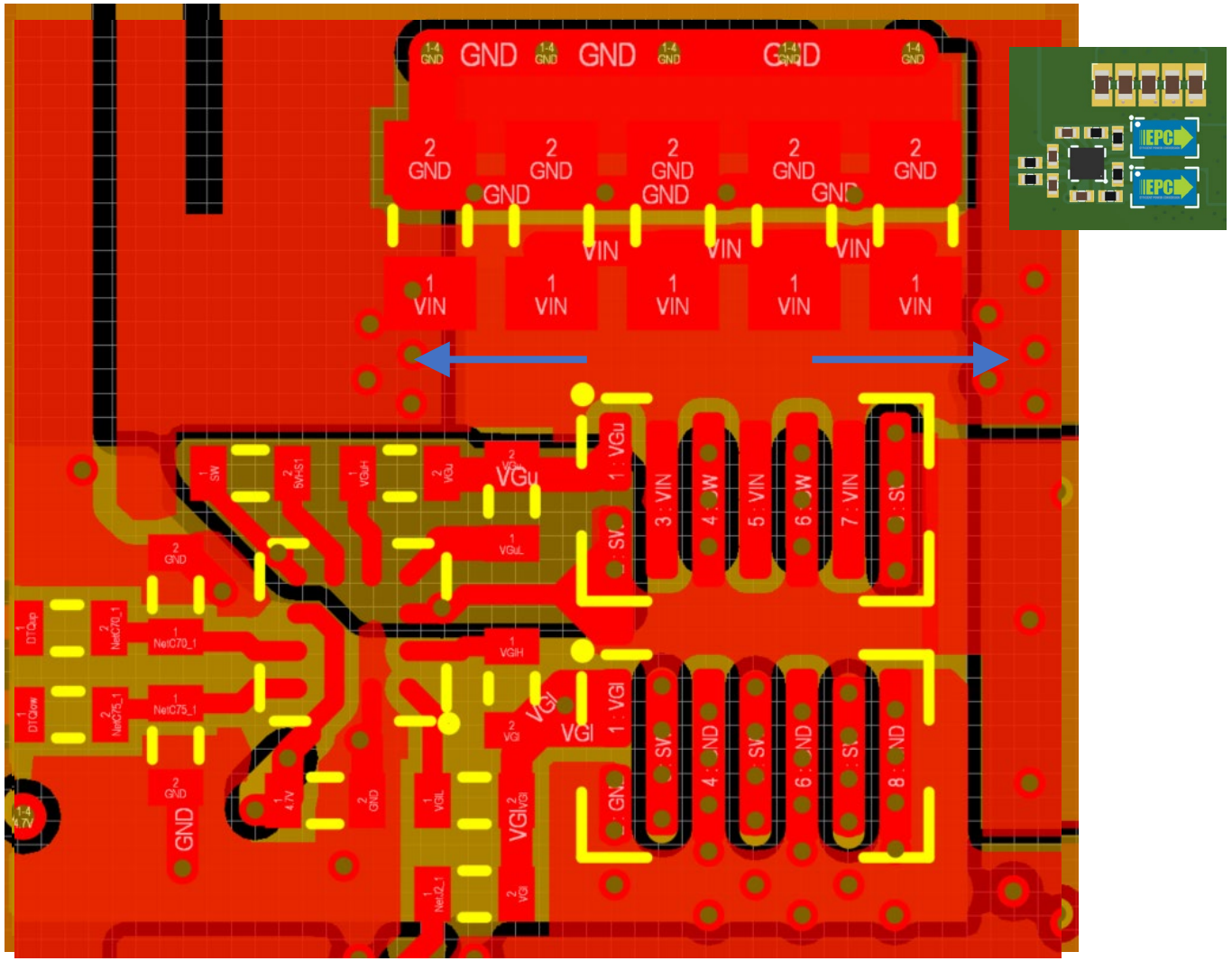
Fixing Mistakes in Gate Layout

- Pad 2 used ~~only~~ as gate return Kelvin connection **and power loop source**
- **Wider** tracks for gate drive and return (8→15 mils)
- Gate drive return in top layer ~~only~~ **and Inner 1**
- HS floating gate circuitry over **SW node** polygon



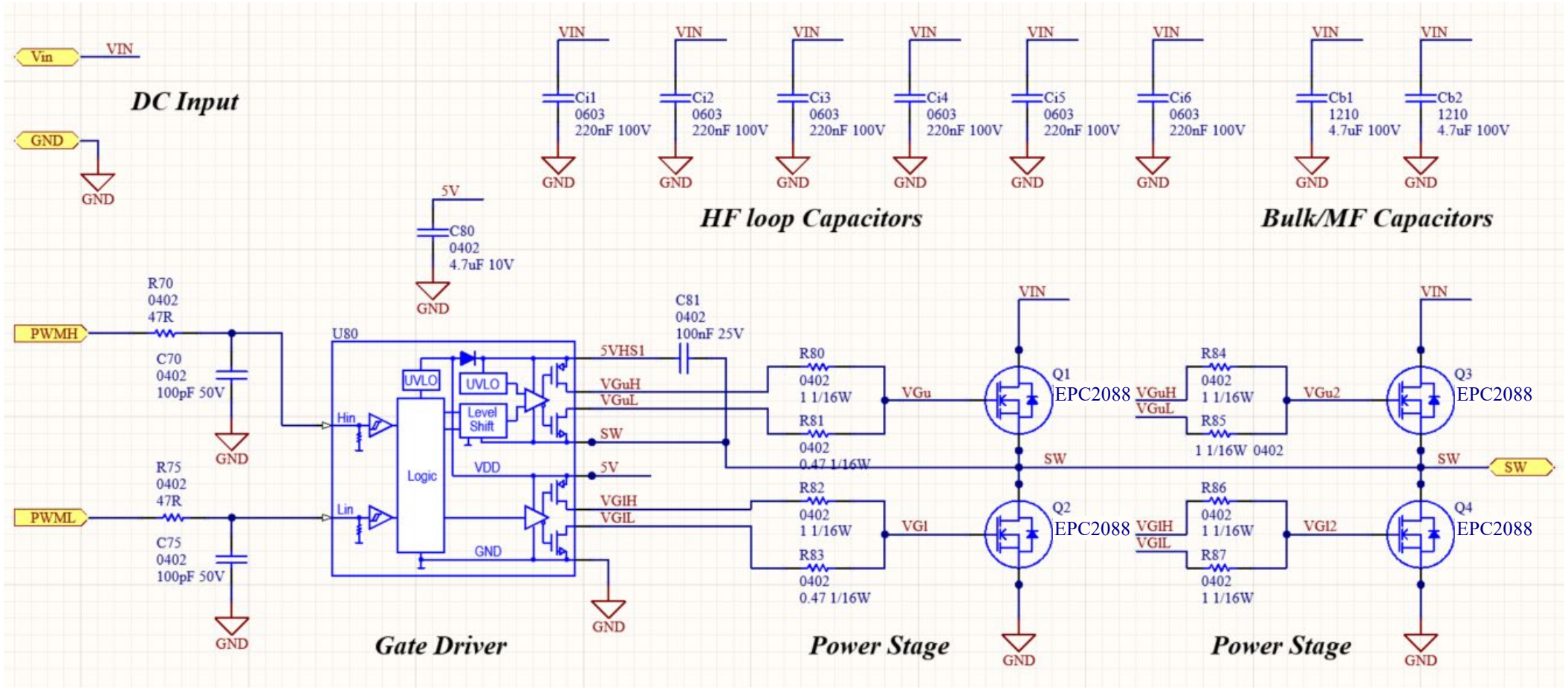
Mistakes in Power Loop

- Vias blocking power loop return
- **First iteration**
- Are those vias in the right position?
- **Second iteration**
  - Capacitors can be moved closer
  - Via in Capacitors' GND Pads can be used



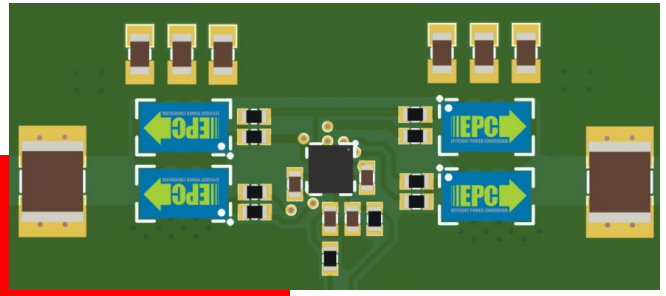
Paralleling FETs for DC/DCs Example

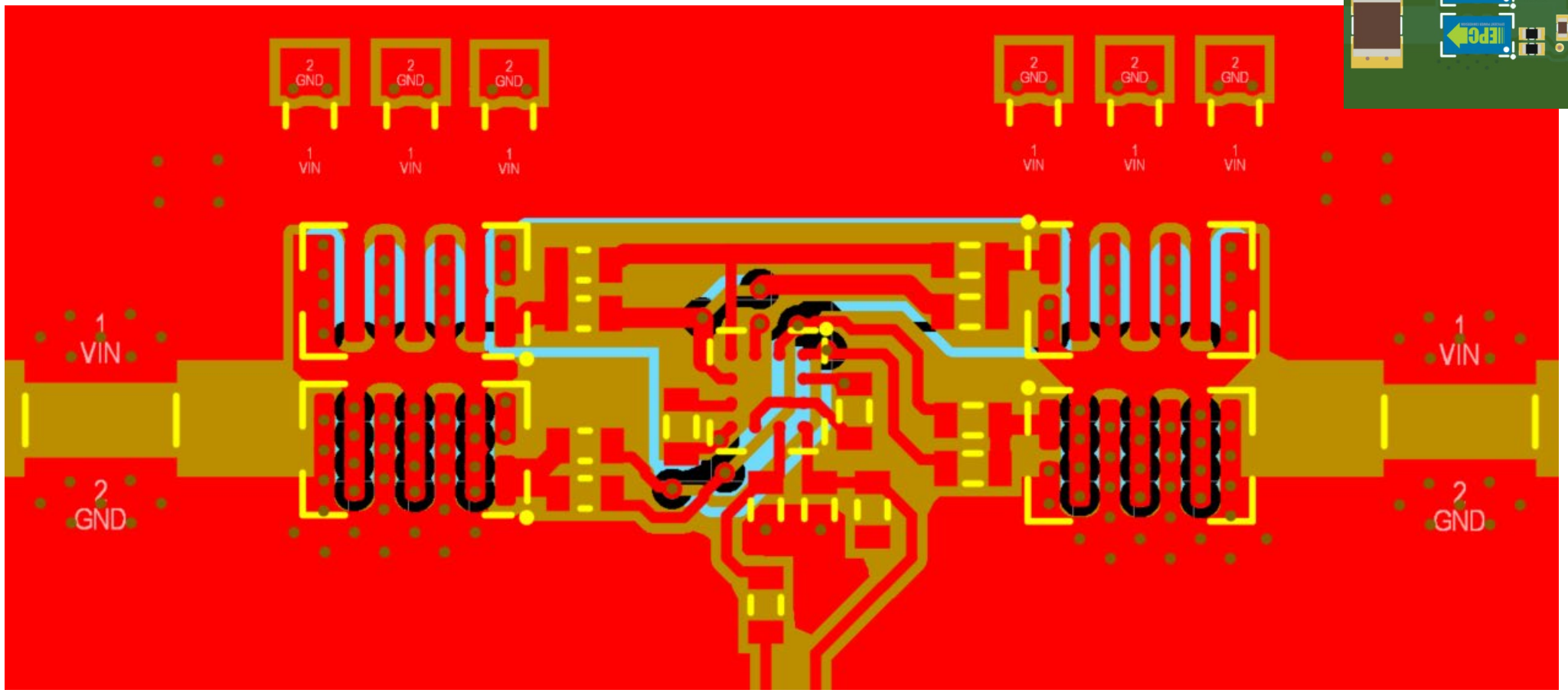
2 // EPC2088 half bridge with LMG1205/MP8699B/LT8418



Paralleling FETs for DC/DCs

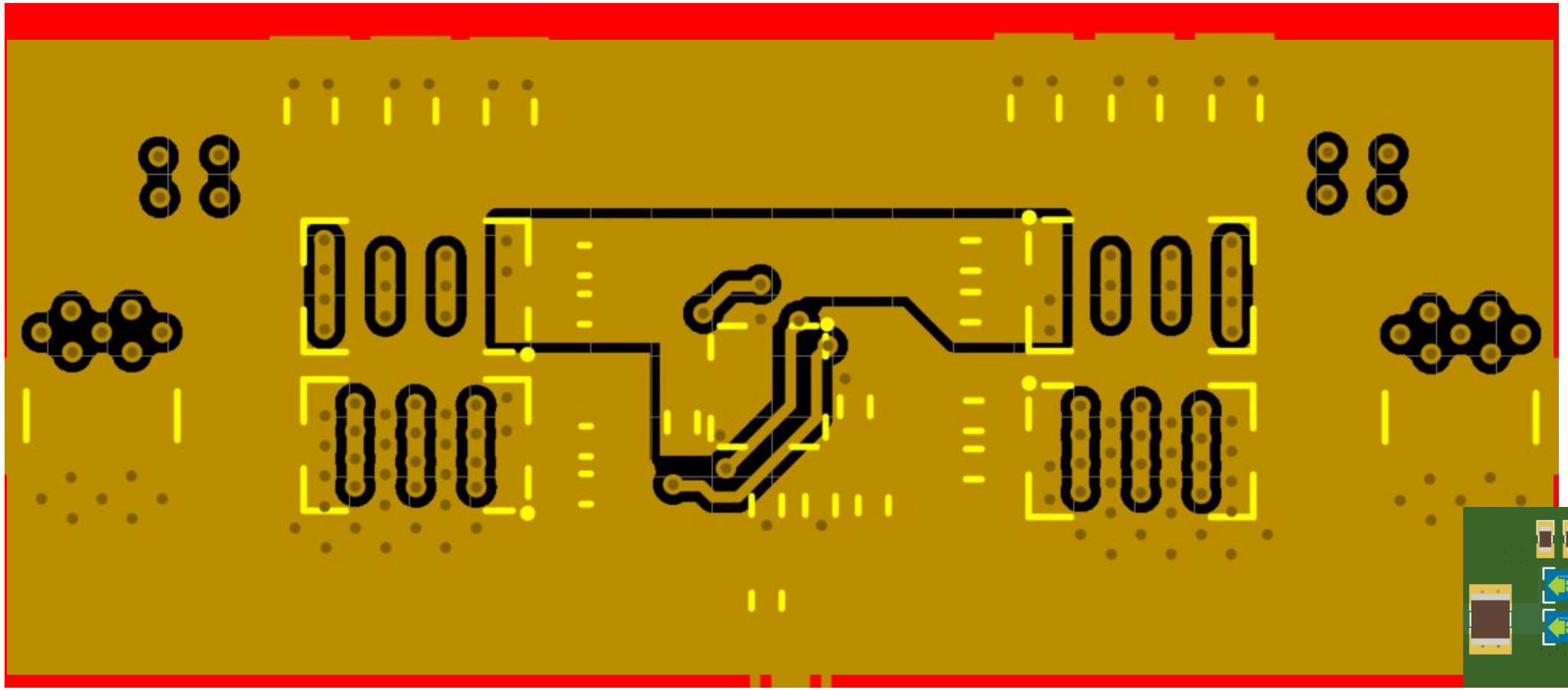
Parallel half bridges: symmetric layout

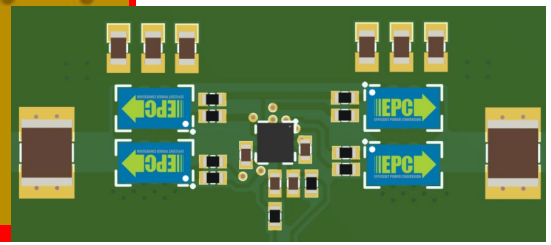




Paralleling FETs for DC/DCs

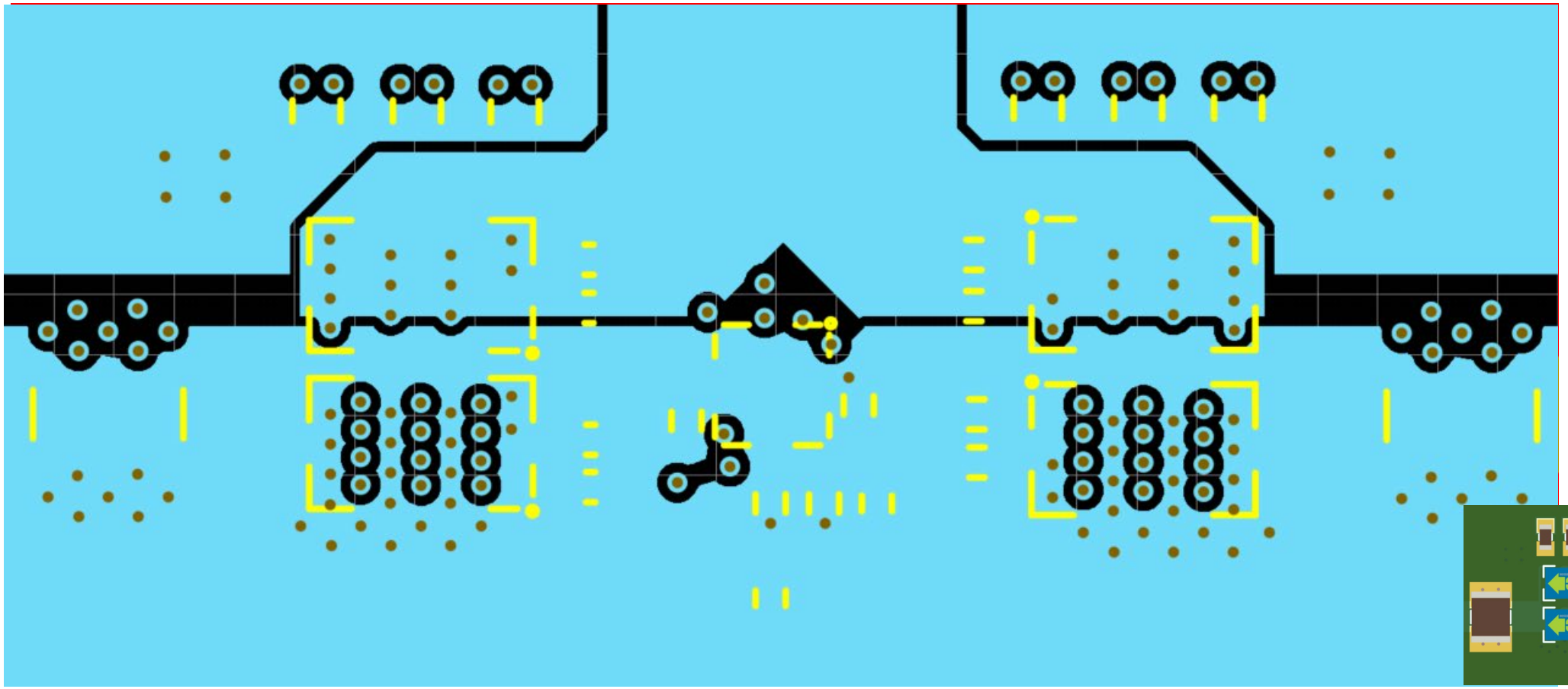
Inner 1 for gate and power loop return (and LS gate routing)

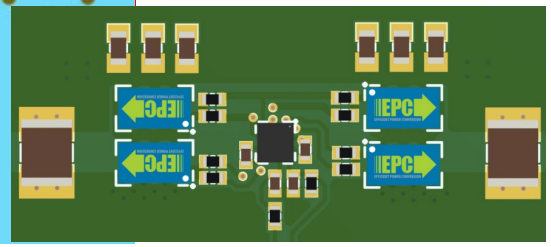




Paralleling FETs for DC/DCs

Inner 3 for LF high current paths (SW node, GND, VBUS)



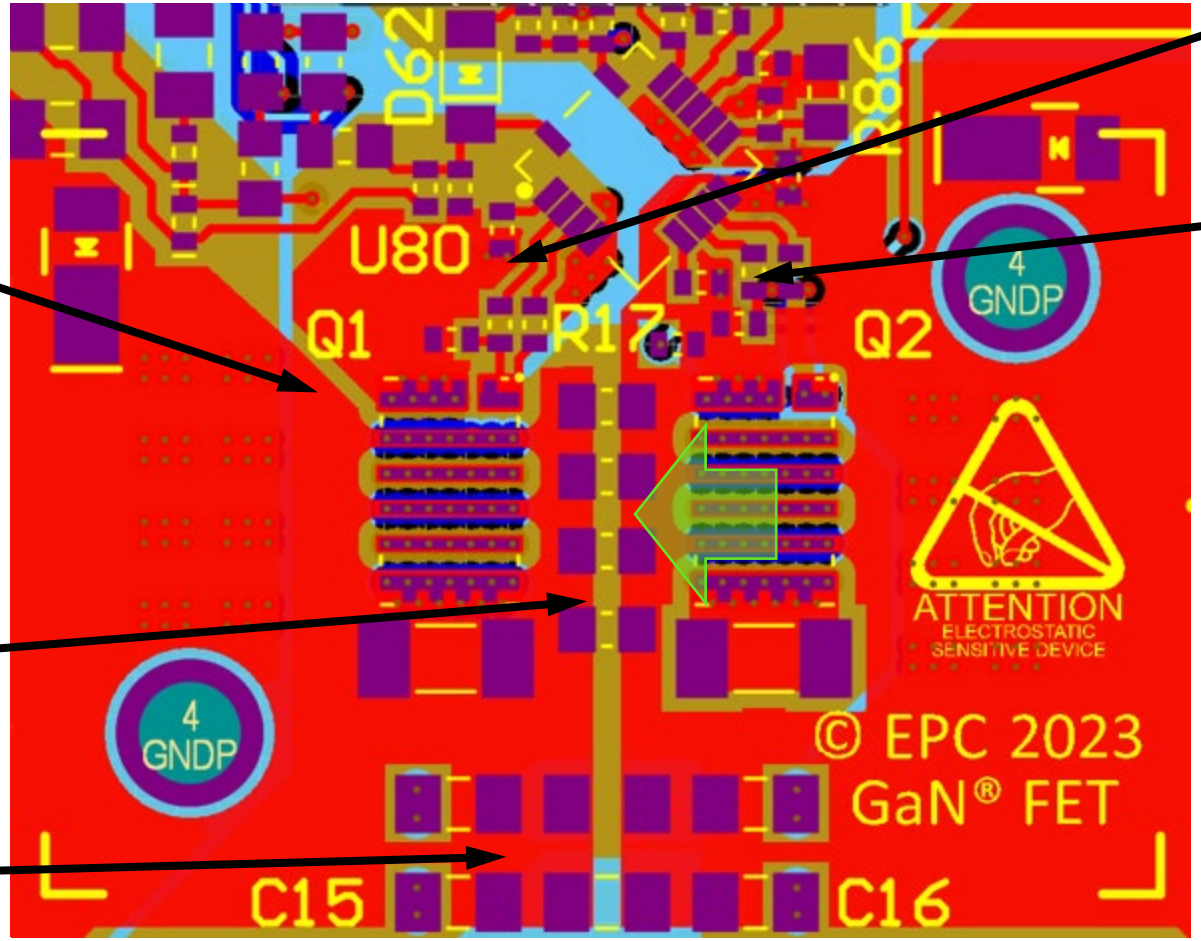


QFN and buried SW Node example

eGaN FET
48 VIAs IN PAD

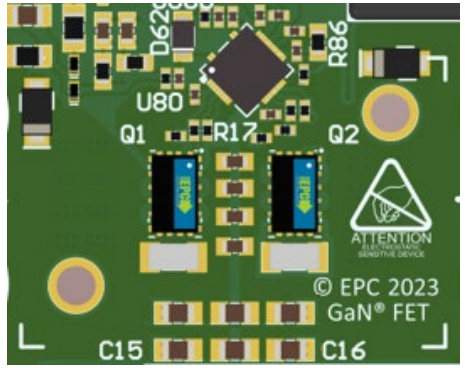
MIDDLE “HF” CAPs

“MF” BYPASS CAPs



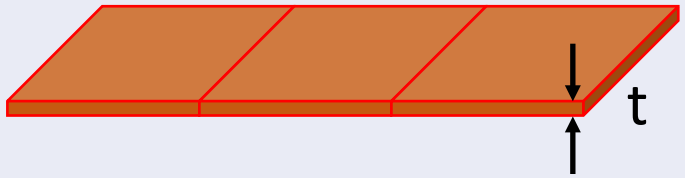
GATE DRIVER (HS)

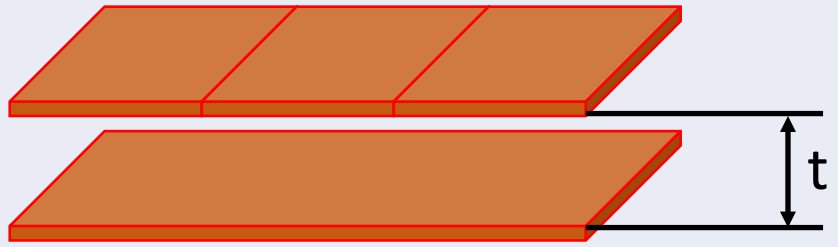
GATE DRIVER (LS)



Reference:
EPC9558P

PCB PARASITICS

PARASITIC	t = 35 µm	t = 70 µm	CONFIGURATION
RESISTANCE PER SQUARE	700 µΩ	350 µΩ	

PARASITIC	t = 0.1 mm	t = 0.2 mm	CONFIGURATION
INDUCTANCE PER SQUARE	130 pH	260 pH	
CAPACITANCE PER AREA	250 pF/in^2 40 pF/cm^2	125 pF/in^2 20 pF/cm^2	

VIAs PARASITICS

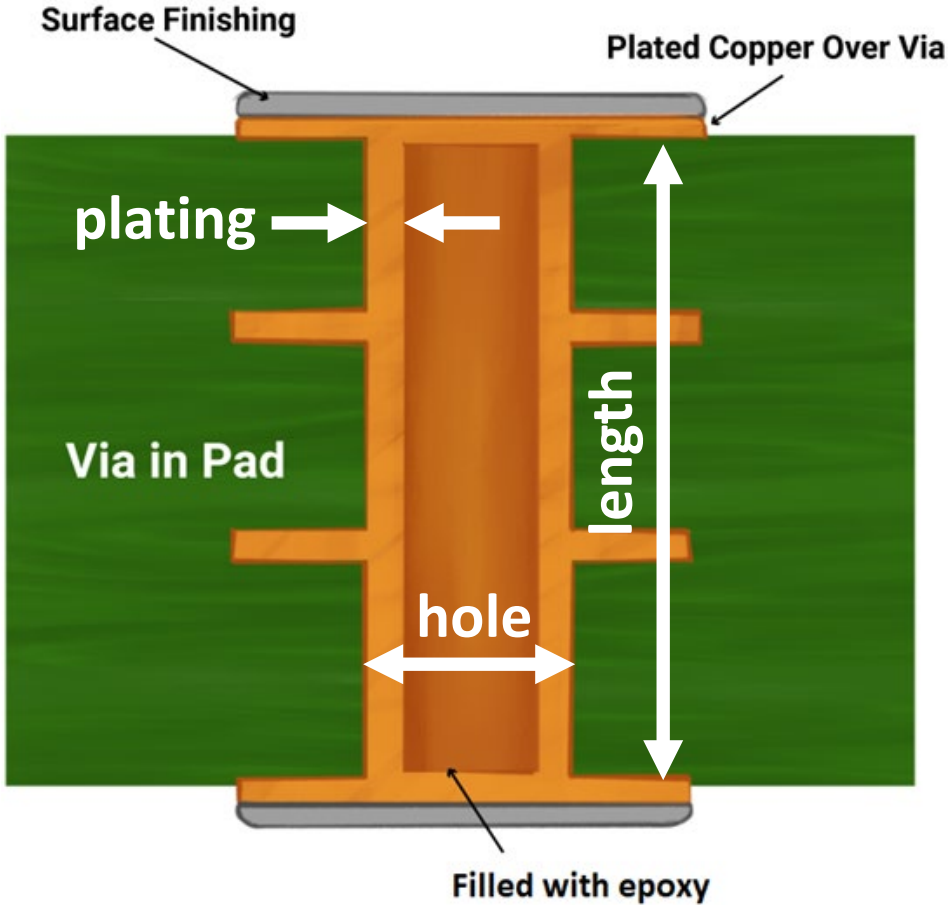
VIA RESISTANCE:

2 mΩ

300 °K/W

- 0.2 mm hole
- 20 µm plating (IPC Class 2)
- **1.5mm length**

Keep it short! Connection to a nearby layer is much better!



VIPPO (Via In Pad Plated Over)

IPC4761 Via Type VII

VIAs PARASITICS

WHAT ABOUT INDUCTANCE?

~1 nH

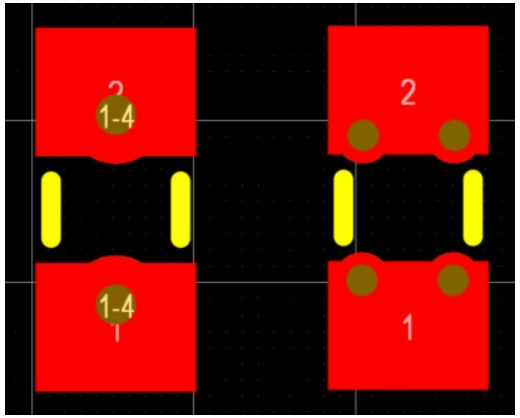
Depends on:

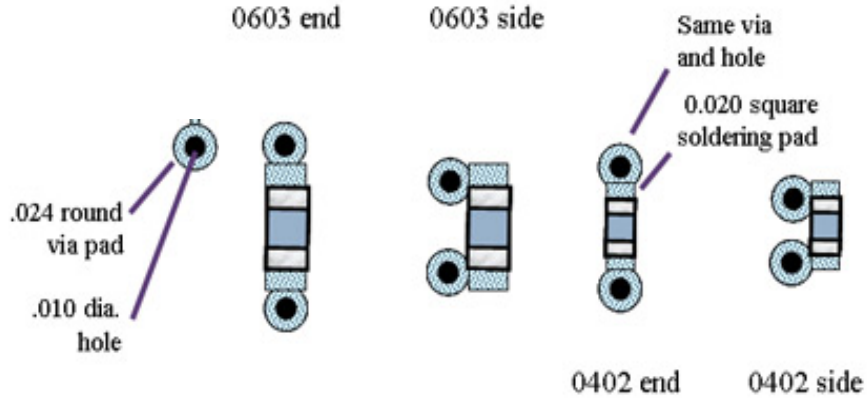
- via placement vs pad
- Diameter & **length**

Keep it short!

VIPPO is better!

0603 VIA IN PAD





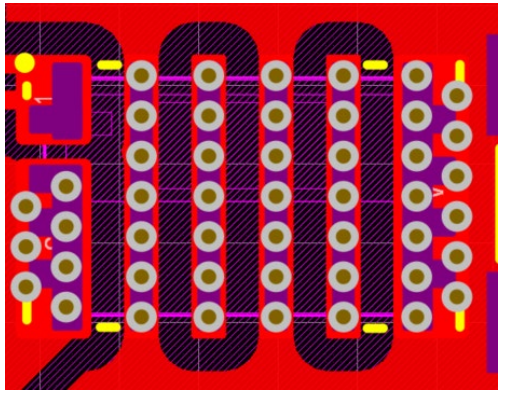
Via length	0603 end	0603 side	0402 end	0402 side
.004	0.50	0.36	0.42	0.26
.006	0.59	0.46	0.50	0.32
.010	0.77	0.61	0.67	0.40
.020	1.16	0.85	1.01	0.60

(nH)

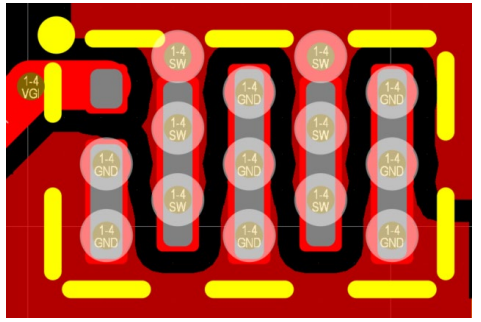
Credits to Howard Johnson, Signal Consulting, Inc.

VIAs & EPC DEVICES

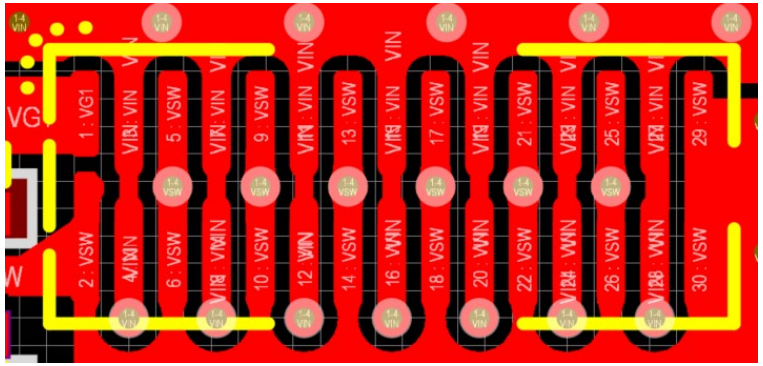
QFN (i.e. EPC2302)
0.2 mm drill, 0.35-0.4 mm annular ring diam. (≤ 2 oz. Cu).
0.5 mm annular ring diam. in inner layers for up to 4 oz.
Up to 0.25 mm / 0.5mm (10/20 mils) possible.



CSP 0.5 mm pitch (i.e. EPC2204, EPC2619, EPC2218):
Goal is 7 mils creepage (Cu pad to Cu pad). Options:
1) vias in source pads only: 0.2mm drill, 0.35mm diam.
2) interleaved (see picture): 0.2mm drill, 0.35mm diam.
3) smaller vias using laser drilling (+$)



CSP 0.4mm pitch (i.e. EPC2206):
Requires 7 mils track width in the notch and 7 mils creepage



Motor drive vs DC/DC

	DC/DC	Motor Drive
Current	5 – 100 A	20 – 400 A
Switching Frequency	> 200 kHz	50 – 200 kHz
dI/dt	5 – 20 A/ns	1 – 10 A/ns
Current transition time	1 – 5 ns	5 – 50 ns
dV/dt	20 – 100 V/ns	2 – 10 V/ns
Voltage transition time	1 – 5 ns	10 – 50 ns
Dead time	5 – 20 ns	20 – 100 ns
Switching losses vs total	30 – 60 %	20 – 40 %

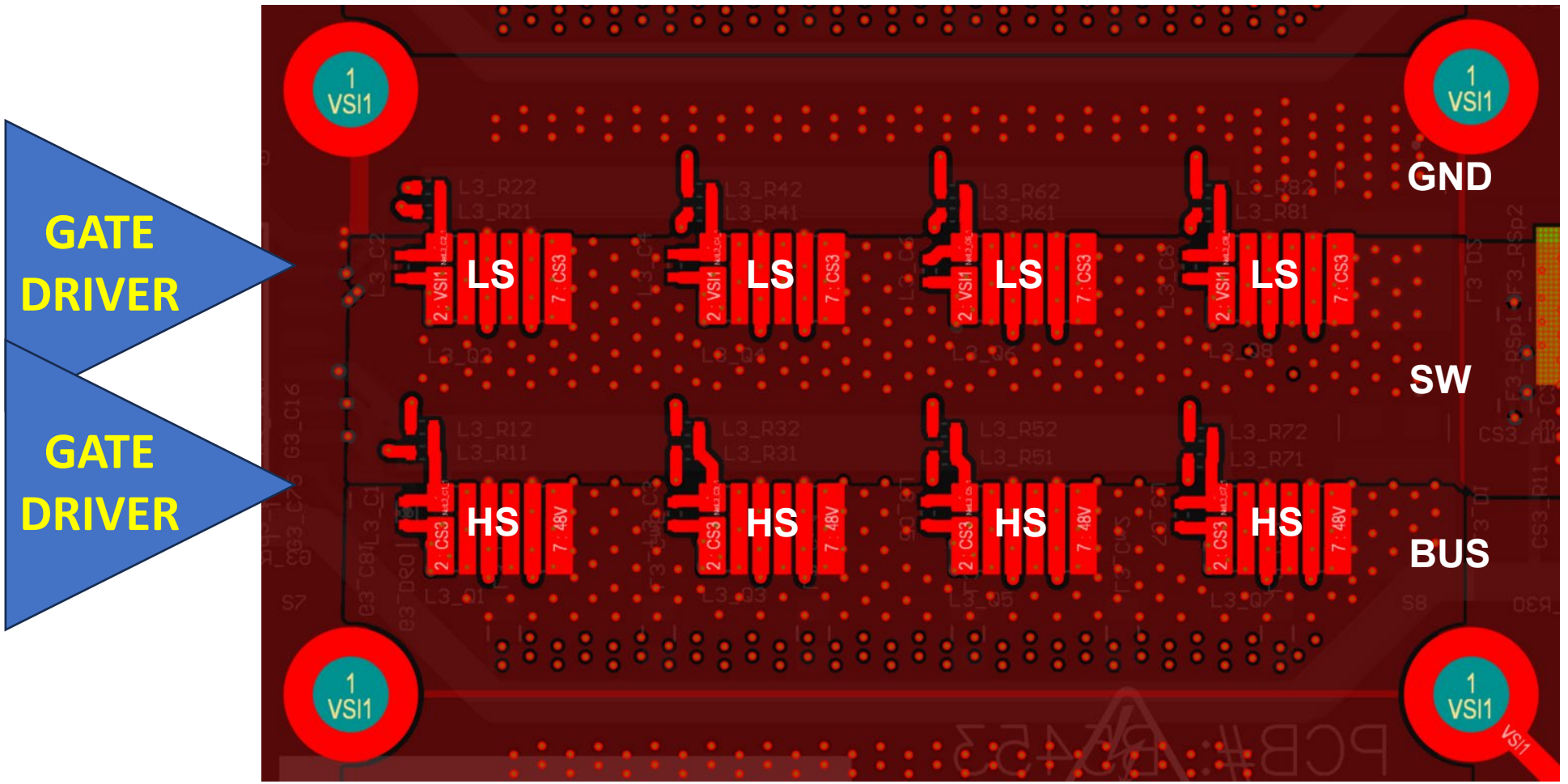
typical values

Layout tips for motor drives

- Local HF capacitors are less critical:
  - they might be omitted
  - interleaving the BUS & GND layers will offer some local distributed capacitance
- Placement of MF/bulk capacitors is less critical:
  - they might be moved where convenient
  - they carry significant ripple current => low resistance path
  - caps and FETs can also be in opposite sides of the PCB
    - external vertical layout is okay
- Distribute gate signals shielding them with source return

Paralleling multiple FETs: EPC9186

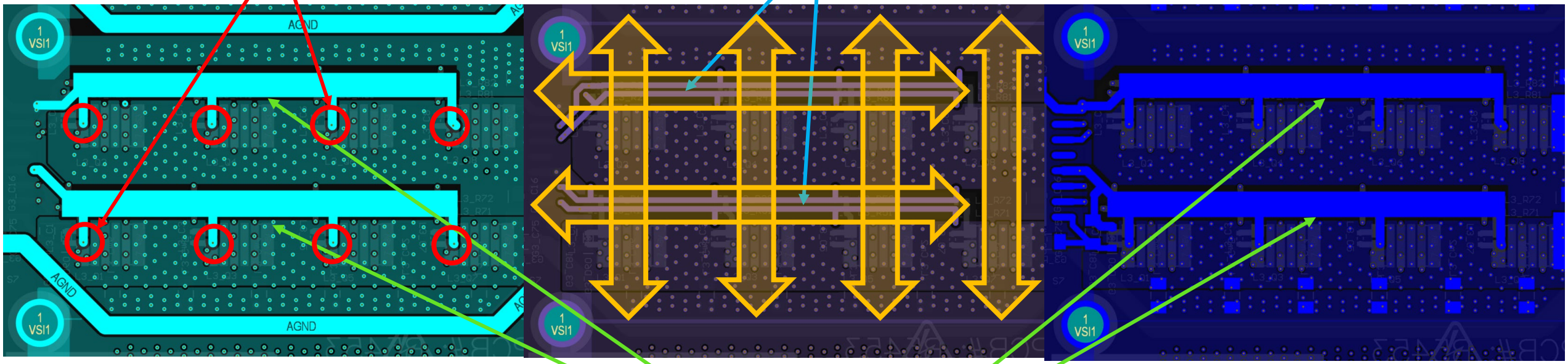
Top side layer



EPC9186 gate drive detail

In-pad vias for source kelvin connection

Shielded gate signal traces (TL)



Mid layer 7

Mid layer 8

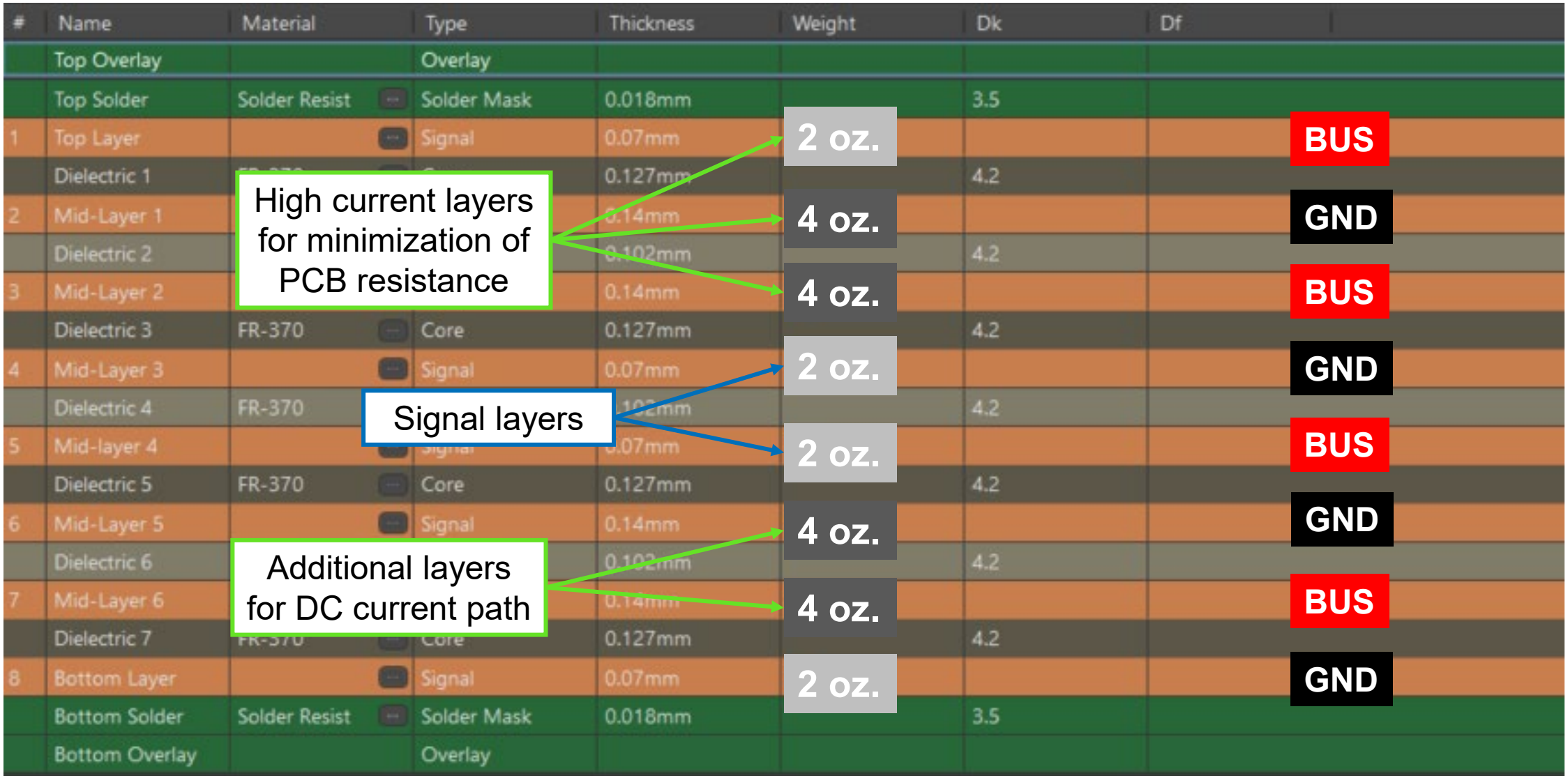
Bottom layer

Source return polygons

More layout tips for high current

- Use a lot of vias: stitching layers will create alternate paths and reduce PCB resistance
- Take care of stackup
- Use copper heavy PCBs: 6 – 10 layers, 2 oz. outer, 4 oz. inner

Example of 8-layer motor drive stackup

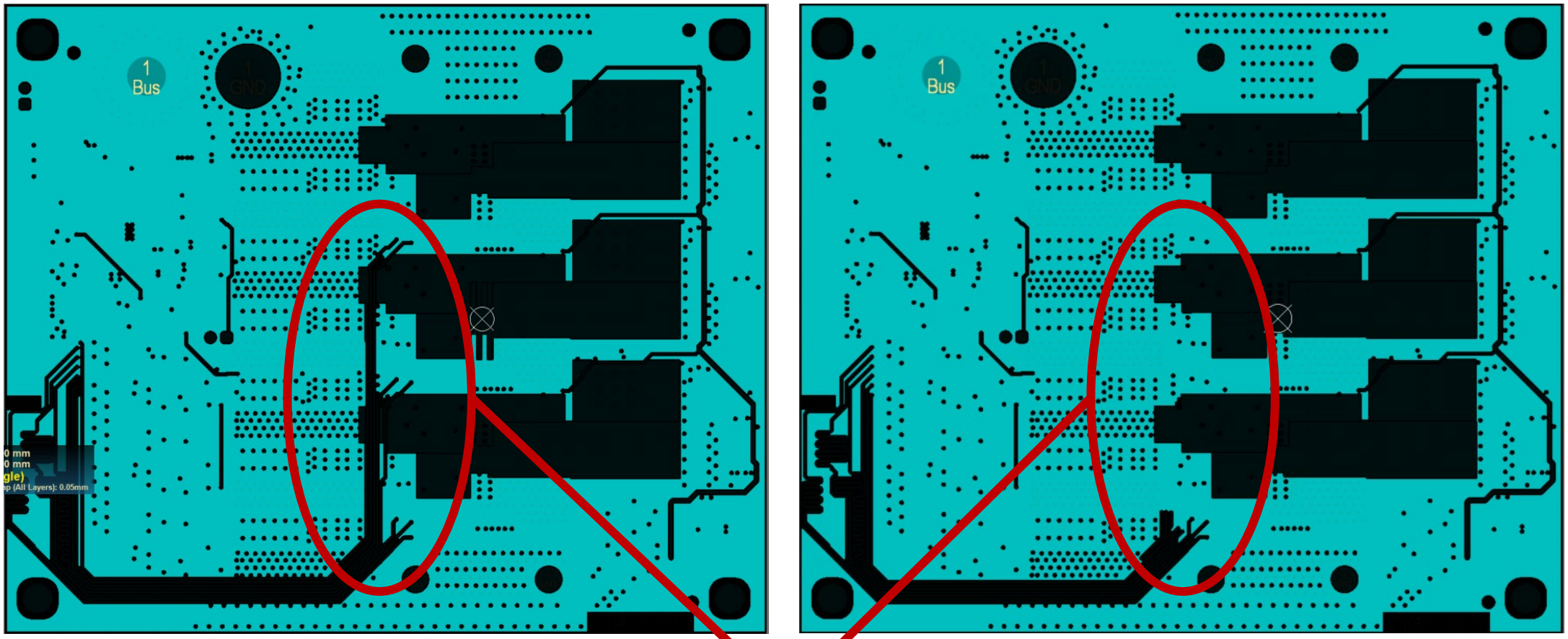


#	Name	Material	Type	Thickness	Weight	Dk	Df	
	Top Overlay		Overlay					
	Top Solder	Solder Resist	Solder Mask	0.018mm		3.5		
1	Top Layer		Signal	0.07mm	2 oz.			BUS
	Dielectric 1			0.127mm		4.2		
2	Mid-Layer 1			0.14mm	4 oz.			GND
	Dielectric 2			0.102mm		4.2		
3	Mid-Layer 2			0.14mm	4 oz.			BUS
	Dielectric 3	FR-370	Core	0.127mm		4.2		
4	Mid-Layer 3		Signal	0.07mm	2 oz.			GND
	Dielectric 4	FR-370		0.102mm		4.2		
5	Mid-layer 4			0.07mm	2 oz.			BUS
	Dielectric 5	FR-370	Core	0.127mm		4.2		
6	Mid-Layer 5		Signal	0.14mm	4 oz.			GND
	Dielectric 6			0.102mm		4.2		
7	Mid-Layer 6			0.14mm	4 oz.			BUS
	Dielectric 7	FR-370	Core	0.127mm		4.2		
8	Bottom Layer		Signal	0.07mm	2 oz.			GND
	Bottom Solder	Solder Resist	Solder Mask	0.018mm		3.5		
	Bottom Overlay		Overlay					

More layout tips for high current

- Use a lot of vias: stitching layers will create alternate paths and reduce PCB resistance
- Take care of stackup
- Use copper heavy PCBs: 6 – 10 layers, 2 oz. outer, 4 oz. inner
- **Avoid tracks or other features creating cuts in polygons**

Layout change – mid layer 2 (Bus plane)

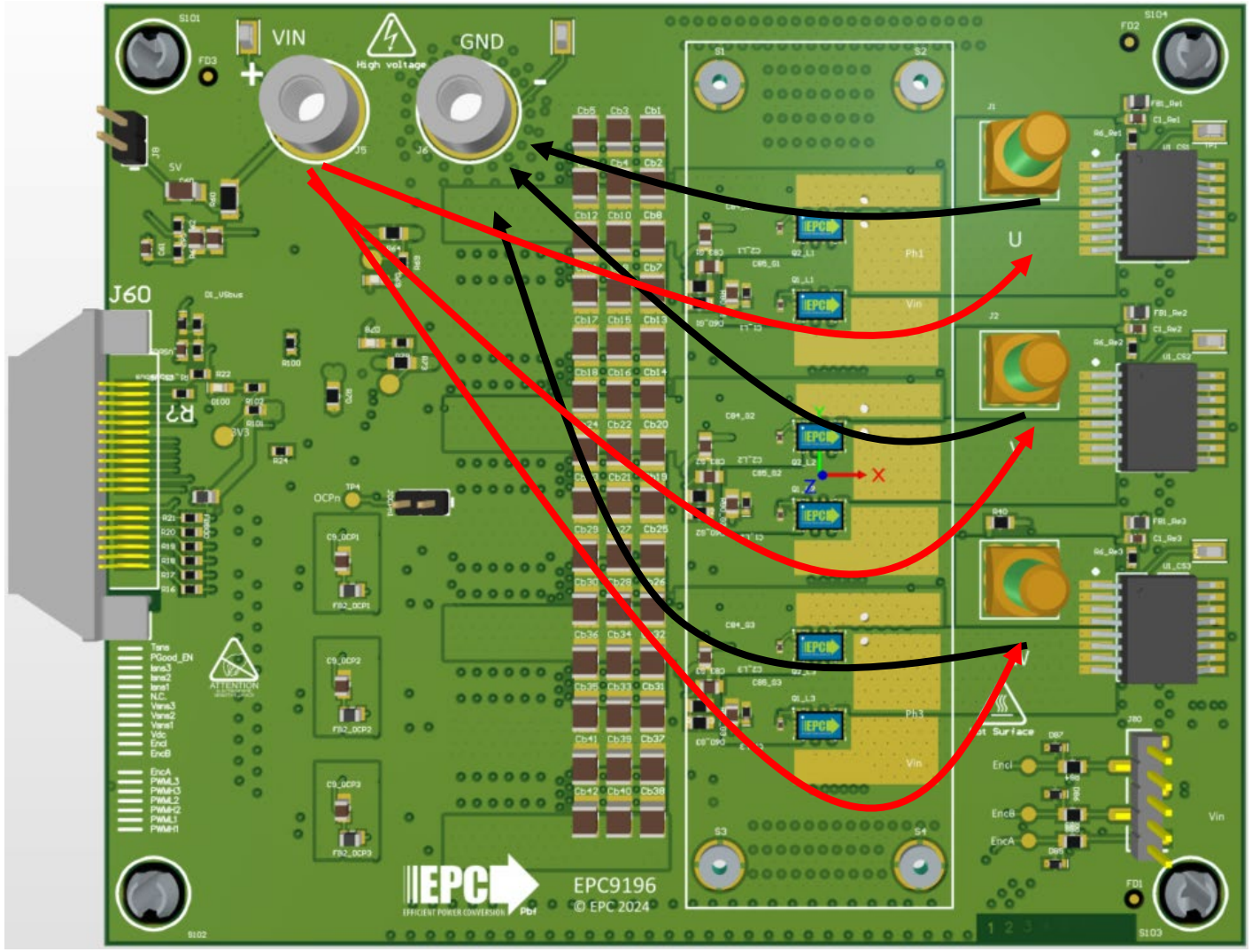


Moved some traces to have a direct path for the Bus polygon

More layout tips for high current

- Take care of stackup
- Use copper heavy PCBs: 6 – 10 layers, 2 oz. outer, 4 oz. inner
- Use a lot of vias: stitching layers will create alternate paths and reduce PCB resistance
- Avoid tracks or other features creating cuts in polygons
- **Use PCB tools to estimate resistance, voltage drop and current density to optimize the layout**
- **Prioritize copper resistance of Vbus → Phase output → GND path instead of "low inductance power loop"**
  - **Switching node carries always the full current**
  - **Vbus and GND are "time shared"**

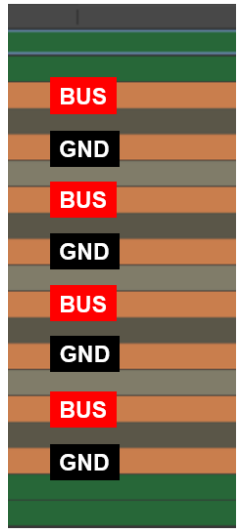
PCB resistance estimation path

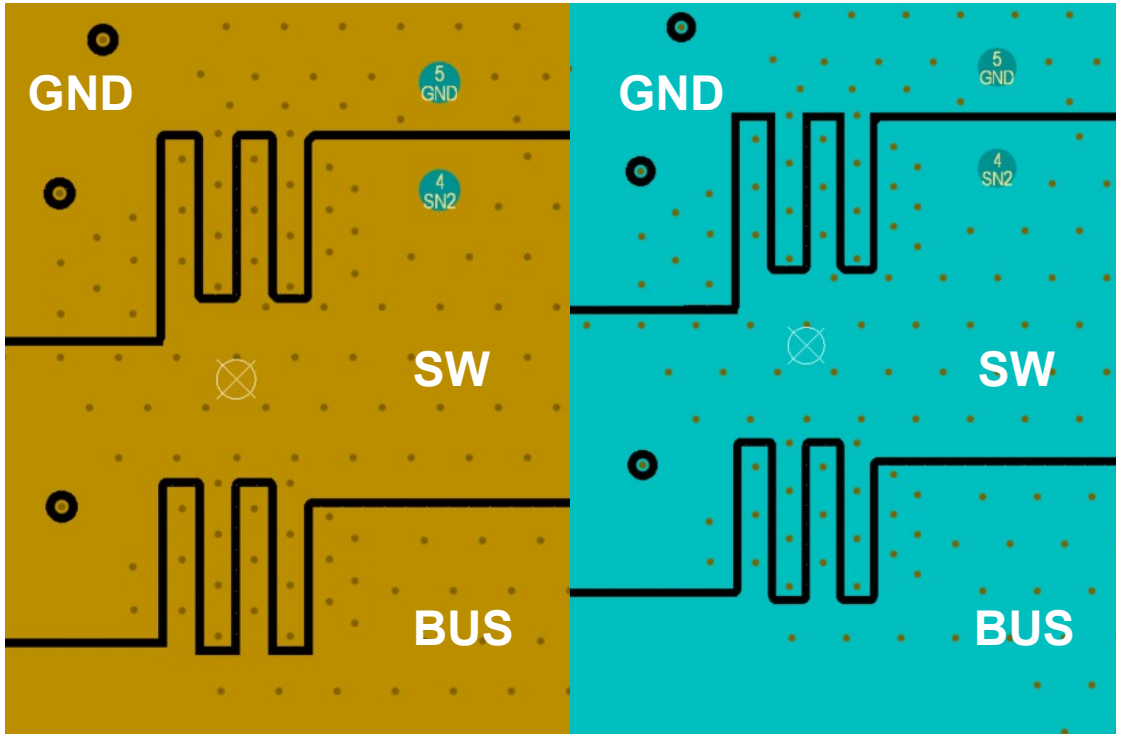


Reference:
EPC91200

Priority to low RDC

- repeat same polygons in all layers….
- …except one for power loop return
- and it can be moved down to, i.e. Inner 3
- or even further down, to minimize via length in DC high current paths





Reference: EPC91200

Summary

- Discussed the 3 important layout-related parasitic inductances
  - Common-Source
  - Power-Loop
  - Gate-Loop
- Presented the best layout designs
  - Internal vertical
  - Parallel FETs
- Provided indications on the PCB design with EPC FETs
- Highlighted priority shift in Motor drive vs DC/DC applications

Additional Resources

- Web page: GaN First Time Right™ Design Process
- App Note: How to Design an eGaN® FET-Based Power Stage with an Optimal Layout
- App Note: Designing PCB Footprint for EPC eGaN® FETs and ICs
- App Note: How to Manually Assemble an eGaN® FET or IC
- App Note: Assembling eGaN® FETs and Integrated Circuits
- Video: Footprint Design – PCB CAD System Independent
- Video: Manual Assembly – eGaN FET Die Attachment Tutorial
- Video: How to GaN 05 – Design Basics: Layout
- Video: Layout Techniques to Maximize GaN Device Performance
- Reliability: Reliability Report - Phase 17

EFFICIENT POWER CONVERSION

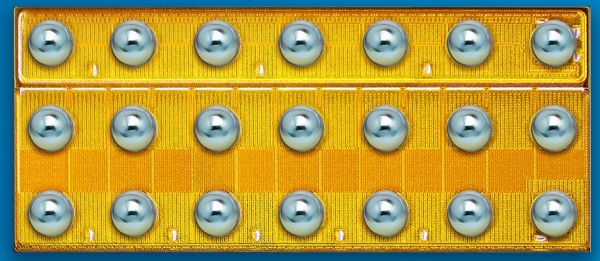
Click on images to learn more



How To GaN Video Series

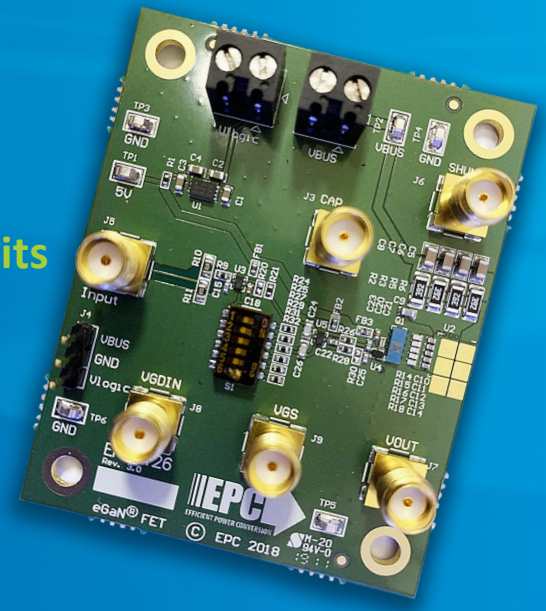


3rd Edition Textbook



eGaN® FETs and ICs

Evaluation Kits



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